
Search for axionlike particle as dark matter candidate with nuclear magnetic resonance

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Declaration of Authorship

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Notes

To ensure clarity and easy connection between the theoretical and experimental parts, constants such as light speed and Planck constant are explicitly written in this thesis. Hope this thesis can be a useful reference for future CASPEr work.

Abstract

Axions and axionlike particles (ALPs) are well-motivated candidates for the dark matter that makes up most of the matter content of the universe. In the nuclear-gradient interaction, an ALP dark-matter field acts as an effective oscillating pseudomagnetic field on nuclear spins, and nuclear magnetic resonance (NMR) provides a resonant method for detecting this weak drive.

This thesis develops the signal models, frequency-scanning strategy, and data-analysis framework for ALP searches with the Cosmic Axion Spin Precession Experiment (CASPER), and applies them to proof-of-principle measurements with the CASPER-Gradient-LowField apparatus.

Two ALP dark-matter models are considered: a virialized Milky Way halo, which gives a stochastic signal with finite coherence time and a linewidth set by the velocity distribution, and a halo of ALPs gravitationally bound to Earth, which can produce long-lived, narrowband, and direction-dependent signals with potentially enhanced local dark-matter density.

For frequency scans, 2 regimes are identified according to the relative magnitudes of the NMR coherence time T_2^* and the ALP coherence time τ_a . In the long-measurement limit, the coupling sensitivity improves only as $T_{\text{tot}}^{-1/4}$ with total integration time in both regimes. To maximize the sensitivity, the relaxation time T_2^* should be as long as possible, and the frequency step size should equal the sensitive bandwidth of a single measurement.

The experimental chapter describes the CASPER-Gradient-LowField apparatus used in this thesis, including the superconducting magnet system, the methanol sample, the pickup coil, the SQUID detection chain, and the Python-based control and acquisition software. The sample consists of thermally polarized liquid methanol at about 180 K, with a proton polarization of only 1.8×10^{-7} , which is the dominant limitation on sensitivity.

The data-analysis pipeline includes time-series sanity checks, Fourier transformation, filter correction, baseline removal, normalized-power-excess thresholding, and outlier validation. The Milky Way ALP search scanned a band of

$$1.348\,570\,\text{MHz} \pm 120\,\text{Hz}$$

in three sequential scans with 100 s acquisitions at each frequency step. No statis-

tically significant candidate survived the cross-scan consistency checks, yielding a 95 % confidence-level limit of $|g_{\text{ap}}| \lesssim 10^{-2} \text{ GeV}^{-1}$ over the scanned mass range of

$$5.577\,237 \text{ neV}/c^2 \pm 496 \text{ feV}/c^2.$$

The Earth-bound ALP search scanned

$$1.348\,575 \text{ MHz} \pm 24 \text{ Hz}$$

with a measurement duration of 1 h at each of 8 frequency steps. No statistically significant candidate was found, giving an approximate 95 % confidence-level limit of $|g_{\text{ap}}| \lesssim 10^{-9} \text{ GeV}^{-1}$ over the scanned mass range of

$$5.577\,258 \text{ neV}/c^2 \pm 99 \text{ feV}/c^2$$

under the Earth-bound halo assumptions used in the analysis.

These measurements demonstrate the use of liquid-state NMR to search for ALP-proton gradient coupling and establish an experimental and analytical basis for more sensitive searches with hyperpolarized samples across different ALP halo models.

The most promising upgrades are hyperpolarized samples, lower noise, better field homogeneity, and larger sample volume. Hyperpolarized methanol could improve the projected coupling sensitivity by orders of magnitude, while highly polarized liquid ^{129}Xe with long transverse relaxation times offers a further route to improved sensitivity. The numerical Bloch-dynamics simulator and statistical framework developed here support the analysis of diverse dark-matter models. These developments could broaden the reach of CASPER to axion streams, dark photons, and other new-physics scenarios.

Zusammenfassung

Axionen und axionartige Teilchen (ALPs) sind gut motivierte Kandidaten für die dunkle Materie, die den größten Teil der Materie im Universum ausmacht. Bei der Gradientenkopplung an Kernspins wirkt ein ALP-Feld der dunklen Materie wie ein effektives oszillierendes pseudomagnetisches Feld auf die Kernspins; die Kernspinresonanz (NMR) bietet eine resonante Methode zum Nachweis dieser schwachen Anregung.

Diese Dissertation entwickelt die Signalmodelle, die Strategie für Frequenzscans und das Datenanalyseverfahren für ALP-Suchen mit dem Cosmic Axion Spin Precession Experiment (CASPER) und wendet sie auf Messungen zum experimentellen Machbarkeitsnachweis mit der CASPER-Gradient-LowField-Apparatur an.

Zwei Modelle für ALP-Dunkle-Materie werden betrachtet: ein virialisierter Halo der Milchstraße, der ein stochastisches Signal mit endlicher Kohärenzzeit und einer durch die Geschwindigkeitsverteilung bestimmten Linienbreite erzeugt, sowie ein Halo aus gravitativ an die Erde gebundenen ALPs, der langlebige, schmalbandige und richtungsabhängige Signale bei einer möglicherweise erhöhten lokalen Dichte dunkler Materie hervorbringen kann.

Für Frequenzscans werden zwei Regime anhand der relativen Größen der NMR-Kohärenzzeit T_2^* und der ALP-Kohärenzzeit τ_a unterschieden. Im Grenzfall langer Messdauer verbessert sich die Kopplungsempfindlichkeit in beiden Regimen mit der gesamten Integrationszeit lediglich gemäß $T_{\text{tot}}^{-1/4}$. Zur Maximierung der Empfindlichkeit sollte die Relaxationszeit T_2^* möglichst lang sein und die Frequenzschrittweite der empfindlichen Bandbreite einer Einzelmessung entsprechen.

Das experimentelle Kapitel beschreibt die in dieser Dissertation verwendete CASPER-Gradient-LowField-Apparatur, einschließlich des supraleitenden Magnetsystems, der Methanolprobe, der Aufnahmespule, der SQUID-Detektionskette sowie der Python-basierten Steuerungs- und Datenerfassungssoftware. Die Probe besteht aus thermisch polarisiertem flüssigem Methanol bei etwa 180 K mit einer Protonenpolarisation von lediglich 1.8×10^{-7} , welche die Empfindlichkeit maßgeblich begrenzt.

Das Datenanalyseverfahren umfasst Plausibilitätsprüfungen der Zeitreihen, Fourier-Transformation, die Korrektur der Filterantwort, die Entfernung der Basislinie, die Anwendung eines Schwellenwerts auf den normierten Leistungsüberschuss sowie die Validierung von Ausreißern. Bei der Suche nach ALPs im Halo der Milchstraße

wurde das Frequenzband

$$1.348\,570\,\text{MHz} \pm 120\,\text{Hz}$$

in drei aufeinanderfolgenden Scans mit einer Messdauer von 100 s pro Frequenzschritt untersucht. Kein statistisch signifikanter Kandidat bestand die Konsistenzprüfungen zwischen den Scans, woraus sich eine Grenze von $|g_{\text{ap}}| \lesssim 10^{-2} \text{ GeV}^{-1}$ bei einem Konfidenzniveau von 95 % im untersuchten Massenbereich

$$5.577\,237\,\text{neV}/c^2 \pm 496\,\text{feV}/c^2$$

ergibt. Bei der Suche nach erdgebundenen ALPs wurde das Frequenzband

$$1.348\,575\,\text{MHz} \pm 24\,\text{Hz}$$

mit einer Messdauer von 1 h an jedem der acht Frequenzschritte untersucht. Es wurde kein statistisch signifikanter Kandidat gefunden, woraus sich unter den in der Analyse verwendeten Annahmen zum erdgebundenen Halo eine ungefähre Grenze von $|g_{\text{ap}}| \lesssim 10^{-9} \text{ GeV}^{-1}$ bei einem Konfidenzniveau von 95 % im untersuchten Massenbereich

$$5.577\,258\,\text{neV}/c^2 \pm 99\,\text{feV}/c^2$$

ergibt.

Diese Messungen demonstrieren den Einsatz der Flüssigkeits-NMR zur Suche nach der ALP-Proton-Gradientenkopplung und schaffen eine experimentelle und analytische Grundlage für empfindlichere Suchen mit hyperpolarisierten Proben unter verschiedenen ALP-Halomodellen.

Die vielversprechendsten Verbesserungen sind hyperpolarisierte Proben, geringeres Rauschen, eine höhere Feldhomogenität und ein größeres Probenvolumen. Hyperpolarisiertes Methanol könnte die prognostizierte Kopplungsempfindlichkeit um Größenordnungen verbessern, während hochpolarisiertes flüssiges ^{129}Xe mit langen transversalen Relaxationszeiten einen weiteren Weg zu höherer Empfindlichkeit bietet. Der hier entwickelte numerische Simulator der Bloch-Dynamik und das statistische Verfahren unterstützen die Analyse verschiedener Modelle dunkler Materie. Diese Entwicklungen könnten die Suchmöglichkeiten von CASPER auf Axionströme, dunkle Photonen und weitere Szenarien neuer Physik ausweiten.

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Six years have passed since I began my journey in Mainz. It feels like a fleeting moment, but it has also been a long journey of growth and learning. Choosing to dedicate myself to the CASPER project was a wise decision, and I have gained much more than skills and knowledge from it. Now the journey is coming to a temporary end, and I would like to take this opportunity to express my gratitude to all those who have supported me along the way.

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Astrophysical evidence for dark matter comes from galaxy rotation curves, gravitational lensing, large-scale structure, and the cosmic microwave background: visible matter alone cannot account for the observed dynamics. Axions and axionlike particles are theoretically attractive candidates because they are weakly coupled, naturally light, and can generate spin-precession signals accessible to NMR experiments.

This chapter introduces the key ingredients of the search. Section 1.1 reviews the NMR concepts most relevant to the experiment. Section 1.2 summarizes the dark-matter evidence. Section 1.3 describes the axion and ALP framework, including the pseudomagnetic-field analogy that connects ALP physics to NMR. Section 1.4 outlines the CASPER project. The theory of signal estimation and scanning optimization is developed in Chapter 2; the apparatus is described in Chapter 3; measurements and results appear in Chapter 4.

1.1 Nuclear magnetic resonance

Nuclear magnetic resonance is a technique for studying nuclear spins in magnetic fields. In the presence of a bias magnetic field, spin states split by the Zeeman effect, resulting in nuclear polarization. A polarized sample acquires a non-zero net magnetization that precesses at the Larmor frequency. Because the Larmor frequency is proportional to the magnetic field, NMR can be a tool for magnetic field measurement.

1.1.1 Zeeman splitting and polarization

In a bias magnetic field B_0 , a spin-1/2 nucleus splits into two energy levels separated by

$$\Delta E = \mu B_0 = \hbar \gamma B_0, \quad (1.1)$$

where μ is the magnetic moment and γ is the nuclear gyromagnetic ratio. This energy difference sets the Larmor (resonance) frequency

$$\nu_L = \frac{\gamma B_0}{2\pi}. \quad (1.2)$$

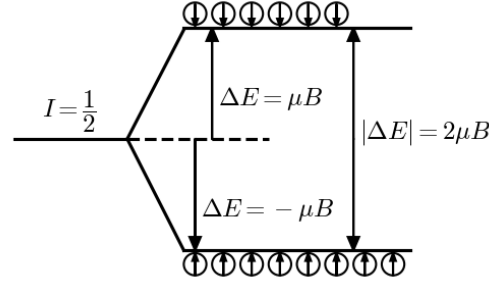


Figure 1.1: Zeeman splitting of a spin-1/2 nucleus in a magnetic field B . The energy difference ΔE between the two spin states is proportional to B .

The splitting is illustrated in Figure 1.1. For protons [44],

$$\gamma = 2\pi \times 42.577\,478\,461(18) \text{ MHz/T}, \quad (1.3)$$

so a bias magnetic field of 31.67 mT, as used in this work, gives $\nu_L \approx 1.348$ MHz.

At thermal equilibrium, the spin populations obey the Boltzmann distribution. In the high-temperature limit $\Delta E \ll k_B T$ (satisfied for all conditions in this thesis), the fractional spin polarization is

$$p \approx \frac{\hbar \gamma B_0}{2k_B T}. \quad (1.4)$$

At $B_0 = 1$ T and $T = 300$ K, the proton thermal polarization is only $p \approx 3 \times 10^{-6}$ [65]. Hyperpolarization techniques—such as spin-exchange optical pumping (SEOP) or parahydrogen-induced polarization (PHIP)—can boost p by several orders of magnitude [26].

1.1.2 Spin-projection noise

Even in the absence of a bias magnetic field, an ensemble of N spins does not cancel perfectly to give a zero net magnetization. Statistical fluctuations leave a residual transverse moment of order $\sqrt{N} \mu$ (where $\mu = \hbar \gamma / 2$ is the nuclear magneton), giving a magnetization noise floor

$$M_{\text{SPN}} \sim \frac{\sqrt{N} \mu}{V}. \quad (1.5)$$

This spin-projection noise (SPN) is the fundamental quantum limit for spin measurements [6, 12, 43]. For a sample of 1 cm^3 of liquid water, there are approximately $N \sim 10^{23}$ protons. SPN gives a magnetization noise floor of $\sim 4 \times 10^{-9} \text{ A/m}$. In comparison to the thermal-equilibrium magnetization of 10^{-8} A/m at 1 T and 300 K, the SPN is one order of magnitude less significant [65]. With lower nuclear polarization or a smaller sample volume, the SPN can become a limiting factor for the sensitivity of the experiment.

1.1.3 Bloch equations

The evolution of the nuclear spin magnetization M in magnetic fields is governed by the Bloch equations [1]:

$$\frac{dM_x}{dt} = \gamma(M \times B_{\text{eff}})_x - \frac{M_x}{T_2}, \quad (1.6)$$

$$\frac{dM_y}{dt} = \gamma(M \times B_{\text{eff}})_y - \frac{M_y}{T_2}, \quad (1.7)$$

$$\frac{dM_z}{dt} = \gamma(M \times B_{\text{eff}})_z - \frac{M_z - M_{\text{eqb}}}{T_1}, \quad (1.8)$$

where B_{eff} is the total effective field, M_{eqb} is the equilibrium magnetization, and T_1 , T_2 are the longitudinal and transverse relaxation times. In most NMR experiments, the bias magnetic field B_0 is much larger than any other applied magnetic field. Therefore, the equations are most efficiently solved in the rotating coordinate frame (RCF), which removes the fast Larmor precession at ν_L in the laboratory frame and exposes the much slower dynamics induced by the additional fields.

1.1.4 Relaxations

Relaxation describes the return of the spin ensemble toward equilibrium after it has been perturbed. In most NMR experiments, a bias magnetic field is applied, which defines the longitudinal direction, and the transverse plane is perpendicular to it¹.

The return of the magnetization to equilibrium along the longitudinal direction is induced by energy exchange with the environment (spin-lattice relaxation) or through spin-spin interactions (spin-spin relaxation). Its rate is characterized by the relaxation time T_1 or relaxation rate $1/T_1$. The “return” can happen in two directions: when the longitudinal magnetization is smaller than the equilibrium

¹ There are NMR experiments where a zero to ultra-low field (ZULF) [10] is applied to study phenomena like the spin-spin interaction, which are difficult to observe at high fields.

value, it grows toward equilibrium; when it is larger (hyperpolarized), it decays. A short T_1 is helpful for rapid repetition of measurements using thermally polarized samples, while for hyperpolarized samples, a long T_1 is preferred to preserve polarization for longer measurements. Its duration can be tuned by adjusting settings such as the bias magnetic field, temperature, and sample composition. For instance, oxygen is a paramagnetic molecule, and dissolved oxygen in the sample can shorten T_1 by providing an additional relaxation channel.

Transverse relaxation, which determines how quickly coherent precession dephases, is characterized by several different time scales, including T_2 , T_δ and T_2^* . T_2 characterizes microscopic spin-spin interactions and T_δ quantifies the dephasing caused by macroscopic magnetic-field inhomogeneity across the sample volume. T_2^* is the effective transverse relaxation time that incorporates both contributions, and it is always shorter than either of the two, as given by the relation

$$\frac{1}{T_2^*} = \frac{1}{T_2} + \frac{1}{T_\delta}. \quad (1.9)$$

The T_2 time can be adjusted by changing the sample composition, while T_δ can be prolonged by improving the field homogeneity through shimming.

For brevity, relaxation processes are referred to by the symbols of their characteristic time scales in this thesis, for example, T_1 relaxation.

1.1.5 Experimental schemes

In a common NMR experiment, the sample is first thermally polarized in a bias magnetic field, then perturbed by excitation pulses to generate a precession signal in the transverse plane, and finally allowed to relax back to equilibrium. The excitation pulses tip the magnetization away from the longitudinal axis. The tipping angle is determined by the pulse duration and amplitude

$$\theta = \gamma B_{\text{exc}} t. \quad (1.10)$$

where θ is the tipping angle, B_{exc} is the amplitude of the excitation field, and t is the pulse duration. The tipping angle is typically $\pi/2$ or π . These are also referred to as 90° and 180° pulses in the NMR literature.

Free decay

After a $\pi/2$ pulse, the magnetization is tipped into the transverse plane and then decays with the relaxation time T_2^* . The resulting free-decay signal has an expo-

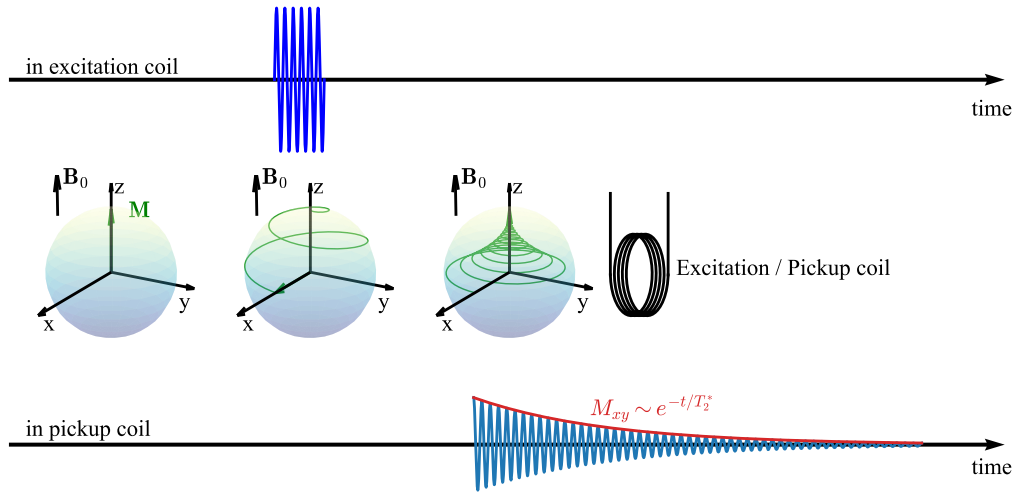


Figure 1.2: Schematic of a free-decay pulsed NMR sequence in the laboratory frame. One excitation pulse tips the magnetization into the transverse plane, where it precesses and decays with time constant T_2^* . The signal is measured by a pickup coil.

nential envelope ($\exp(-t/T_2^*)$) and can be used to measure T_2^* and the Larmor frequency. The process is illustrated in Figure 1.2.

If the measured quantity is the voltage induced in a pickup coil by spin precession, the signal is called free-induction decay (FID). However, in this thesis, the measured quantity is the magnetic flux. Therefore, the more general term “free decay” is used.

Spin echo and CPMG sequence

T_2^* relaxation leads to dephasing of the transverse magnetization due to differences in the Larmor frequency across the sample volume. If a π pulse is applied after a $\pi/2$ pulse, the magnetization is flipped by 180° , and the dephasing is reversed. The reversed dephasing, or “refocusing,” produces a spin echo [33], as illustrated in Figure 1.3. This spin-echo scheme separates spin-spin relaxation from relaxation caused by field inhomogeneity.

The Carr–Purcell–Meiboom–Gill (CPMG) [17, 41] sequence extends the spin-echo scheme by applying a $\pi/2$ pulse followed by a train of π refocusing pulses. Each refocusing pulse produces an echo, and the decay of the echo amplitudes gives a measurement of T_2 . In this thesis, CPMG is used as a pulsed-NMR diagnostic to characterize the sample relaxation before and after measurements.

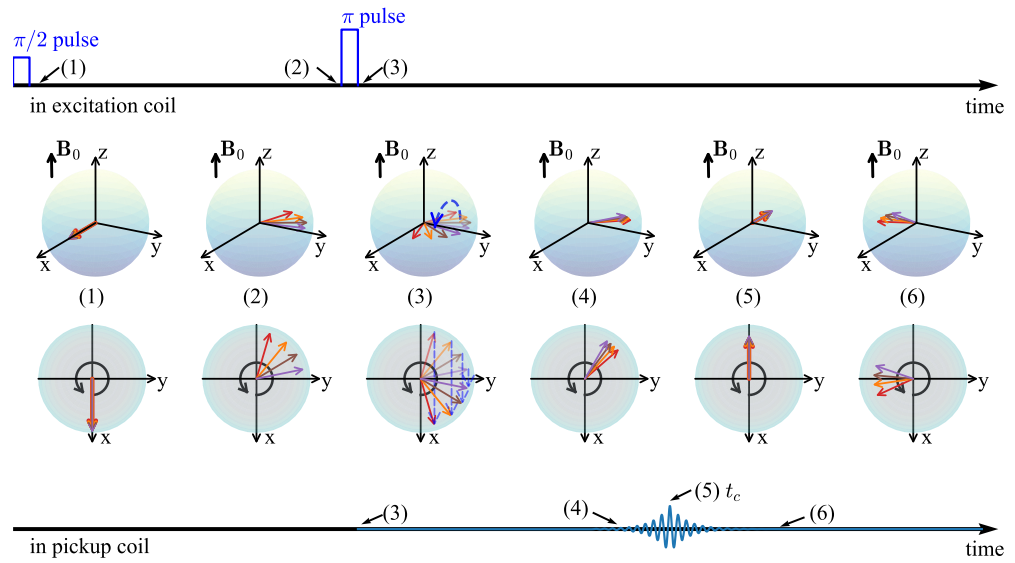


Figure 1.3: Schematic of the spin-echo sequence in the laboratory frame. Black circular arrows indicate the precession. Colored arrows represent the magnetization of small sample packets precessing at different Larmor frequencies. The π pulse reverses dephasing from static field inhomogeneity and produces echoes. The indices in brackets indicate the time sequence of the pulses and echoes.

1.2 Dark matter hypothesis

The prevailing cosmological framework is the Λ cold-dark-matter (Λ CDM) model. It combines a cosmological constant Λ , which describes the accelerated expansion of the universe, with a cold, nonrelativistic, and collisionless dark-matter component. Λ CDM provides a successful common description of the cosmic microwave background, the growth of large-scale structure, galaxy clustering, and the expansion history. Its success establishes a powerful baseline, but it does not identify the physical nature of dark matter, and its assumptions are being challenged on galactic and subgalactic scales.

An alternative to dark matter is modified gravity, most prominently Modified Newtonian Dynamics (MOND), which replaces Newtonian dynamics at very small accelerations rather than introducing unseen matter. MOND was originally proposed to explain flat galaxy rotation curves and can reproduce several galactic-scale regularities with an acceleration scale $a_0 \sim 10^{-10} \text{ m s}^{-2}$ [9, 42, 69]. MOND captures several galactic-scale regularities, but it faces difficulties in explaining galaxy clusters and cosmological data. This thesis aims to test the dark-matter hypothesis, so it does not focus on modified-gravity alternatives.

1.2.1 Astronomical evidence

The concept of dark matter arose from astronomical observations that could not be explained by visible matter and standard gravity alone. Here “visible” refers to baryonic matter that emits or scatters electromagnetic radiation, while “dark” refers to matter that interacts primarily through gravity. Galaxy rotation curves provide a classic example: orbital speeds remain approximately constant far beyond the bright stellar disk, rather than decreasing as expected if most of the mass followed the visible matter [52, 56]. This behavior indicates that galaxies are embedded in extended, non-luminous mass halos.

Evidence at larger scales comes from gravitational lensing and colliding galaxy clusters. In the Bullet Cluster, the dominant gravitational mass inferred from lensing is spatially separated from the baryonic mass, which is mostly hot X-ray-emitting gas [19]. This separation is naturally explained if most of the mass is dark and collisionless during the cluster collision, and disfavors the MOND theory. Independently, the cosmic microwave background, big-bang nucleosynthesis, and the growth of large-scale structure require substantially more gravitating matter than can be supplied by baryons. Cosmological fits assign roughly 5 % of the cosmic energy budget to baryonic matter and about 27 % to dark matter [2].

1.2.2 Recent developments

Despite the large-scale success of Λ CDM, its simplest collisionless cold-dark-matter realization faces open questions on smaller scales. Simulations and observations can differ in the predicted abundance of dwarf satellites and in the inner density profiles and diversity of low-mass halos. These tensions may reflect incompletely modeled baryonic processes, or they may indicate that the dark-matter sector is warmer, wave-like, or self-interacting rather than perfectly cold and collisionless [57].

Recent observations increasingly test these possibilities. Work involving Roberto Abraham and collaborators identified the ultra-diffuse galaxy NGC 1052–DF2 as having an unusually small dark-matter content [22], referred to as “dark matter deficiency”. Follow-up work proposed that this galaxy and related objects form a trail produced by a high-speed “bullet-dwarf” collision that separated stars and gas from their original dark-matter halos [23]. Such systems probe the normally tight connection between galaxies and their halos and may help clarify how baryonic and dark matter can become spatially separated. They challenge simple galaxy–halo relations, although collision-driven formation scenarios may accommodate them within the broader Λ CDM framework.

The central structures of halos provide a more direct test of the collisionless-CDM assumption. In self-interacting dark matter (SIDM) models, scattering between dark-matter particles redistributes energy and changes the central density profiles of halos. A recent strong-lensing analysis led by Simona Vegetti found an unusually compact perturber whose inferred structure, if dark-matter dominated, is difficult to reconcile with conventional cold or warm dark matter [50]. One possible interpretation is an SIDM halo that underwent gravothermal core collapse and formed a central black hole [57]. This interpretation is not yet unique, but it highlights the challenges facing Λ CDM at small scales.

1.2.3 Dark matter models

The observations establish the gravitational role of dark matter but do not identify its microscopic nature. In this sense, Λ CDM specifies the required large-scale behavior of the dominant matter component rather than a unique particle. Proposed realizations cover an enormous range of mass and behavior, and several alternatives modify the small-scale predictions of minimal cold dark matter. At the low-mass end, fuzzy dark matter consists of bosons with masses $\sim 10^{-22}$ eV/ c^2 ; their kiloparsec-scale de Broglie wavelengths make wave effects relevant to galactic structure [35]. Other particle candidates include sterile neutrinos, warm dark

matter (WDM), weakly interacting massive particles (WIMPs), and self-interacting particles. At the opposite, macroscopic end, primordial black holes (PBHs) formed in the early universe could contribute some or all of the dark matter within the remaining observationally allowed mass windows [16]. Laboratory searches consequently range from particle-collision and scattering experiments to gravitational surveys and precision measurements [37].

Among all these dark matter models, axions and axionlike particles occupy a particularly interesting part of this landscape. Their low masses permit a coherent, wave-like dark-matter field, while their weak non-gravitational couplings make laboratory searches possible. The following section introduces their origin, field properties, and coupling to nuclear spins.

1.3 Axions and axionlike particles

1.3.1 Axion

Why does quantum chromodynamics (QCD) appear to conserve CP (C and P are the charge conjugation and parity operators, respectively) so precisely even though the theory allows a CP -violating term that should otherwise be observable? This is the strong CP problem. In 1977, Peccei and Quinn proposed a solution to the strong CP problem by introducing a new global $U(1)$ symmetry, now called the Peccei–Quinn symmetry [47, 48]. The symmetry is spontaneously broken at a high energy scale f_a , which generates a pseudo-Goldstone boson called the axion. The potential of the axion field is minimized at the CP -conserving point, naturally explaining why the strong interaction conserves CP . The axion acquires a mass $m_a \sim \Lambda_{\text{QCD}}^2/f_a$ from QCD dynamics. To distinguish this axion model from other axionlike particles, it is also referred to as the “QCD axion” or “PQ-axion” in the literature. Cosmological production via the misalignment mechanism constrains $f_a \lesssim 10^{12}$ GeV (axion mass $\gtrsim 6$ μeV) if the axion is to constitute all dark matter [32].

1.3.2 Axionlike particles

Axionlike particles (ALPs) emerge when a global symmetry is broken [21, 38, 53, 60, 61, 70]. ALPs share the oscillatory and weakly coupled character of the QCD axion, but their mass range can be much wider, since the mechanism of ALP production differs from that of the QCD axion. The ALP mass range has been investigated using astronomical observations [39] and theoretical studies [70]. However, in this

thesis, the ALP mass is regarded as a free parameter, and the search is performed in a range to which the experiment is sensitive.

Wave nature

A nonrelativistic ALP with mass m_a and velocity $\mathbf{v}_a$ ($v_a \ll c$) has energy $E_a \simeq m_a c^2 + \frac{1}{2} m_a v_a^2$ and a Compton frequency $\nu_a = E_a/h$, where h is Planck's constant and c is the speed of light in vacuum. Its de Broglie wavelength is

$$\lambda_{\text{dB}} = \frac{h}{m_a v_a}. \quad (1.11)$$

For an ultralight ALP, this wavelength can be macroscopic ($\gtrsim 100 \mu\text{m}$). Assuming axionlike dark matter is cold, its velocity is expected to be smaller than the escape velocity of the Milky Way halo $v_{\text{esc}} = 550 \text{ km/s}$ [27]. The minimum de Broglie wavelength is $\lambda_{\text{dB}} \sim 7 \times 10^{-5} \text{ m}$ for ALPs in the Milky Way ($m_a \leq 10 \text{ eV}/c^2$) [65]. For the ALP mass range relevant to this thesis ($m_a \leq 2.5 \mu\text{eV}$), the de Broglie wavelength is $\lambda_{\text{dB}} \geq 270 \text{ m}$, much larger than the size of the laboratory. Therefore, the amplitude of the ALP field can be approximated as uniform over the laboratory.

The ALP field of a single free ALP mode can be written as a real field:

$$a(\mathbf{r}, t) = \sqrt{\frac{\hbar^3 c}{2m_a}} (\Psi(\mathbf{r}, t) + \Psi^*(\mathbf{r}, t)) \quad (1.12)$$

$$= \sqrt{\frac{2\hbar^3 c}{m_a}} \Re[\Psi(\mathbf{r}, t)], \quad (1.13)$$

where Ψ is the wavefunction of a single ALP. Consider an ensemble of N_a ALPs with local energy density ρ_a . Its number density is

$$n_a = \frac{\rho_a}{m_a c^2}, \quad (1.14)$$

and the mean occupation number within a de Broglie volume is approximately

$$N_a \simeq n_a \lambda_{\text{dB}}^3 = \frac{\rho_a h^3}{m_a^4 v_a^3 c^2}. \quad (1.15)$$

If the ALP is the dominant dark-matter component, ρ_a is close to the local dark-matter energy density $\rho_{\text{DM}} \sim 0.4 \text{ GeV}/\text{cm}^3$ [27]. For ALP masses relevant to this thesis, $m_a \leq 2.5 \mu\text{eV}$ and velocities $v_a \leq v_{\text{esc}} = 550 \text{ km/s}$, the occupation number

is $N_a \geq 3 \times 10^{27}$, much larger than unity [65]. Therefore, what is measured in the experiment is the collective effect of many ALPs instead of rare single-particle events.

For N_a ALPs that are collectively described by a wavefunction normalized as $\int |\Psi|^2 d^3r = 1$, the associated scalar field is the appropriately scaled real part of the wavefunction, as in Equation (1.13):

$$a(\mathbf{r}, t) = \sqrt{\frac{2N_a \hbar^3 c}{m_a}} \Re[\Psi(\mathbf{r}, t)] . \quad (1.16)$$

The local, time-averaged energy density is $\rho_a = m_a c^2 N_a \langle |\Psi(\mathbf{r}, t)|^2 \rangle_t$, where $\langle |\Psi(\mathbf{r}, t)|^2 \rangle_t$ denotes the time average. Eliminating N_a gives

$$\begin{aligned} a(\mathbf{r}, t) &= \sqrt{\frac{2\hbar^3 \rho_a}{m_a^2 c}} \frac{\Re[\Psi(\mathbf{r}, t)]}{\langle |\Psi(\mathbf{r}, t)|^2 \rangle_t^{1/2}} \\ &= a_0 \frac{\Re[\Psi(\mathbf{r}, t)]}{\langle |\Psi(\mathbf{r}, t)|^2 \rangle_t^{1/2}} , \end{aligned} \quad (1.17)$$

where

$$a_0 = \sqrt{\frac{2\rho_a \hbar^3}{m_a^2 c}} . \quad (1.18)$$

is the local field amplitude. If all N_a ALPs occupy the same mode, the corresponding local field takes the plane-wave form

$$a(\mathbf{r}, t) = a_0 \cos(2\pi\nu_a t - \mathbf{k} \cdot \mathbf{r} + \phi) , \quad (1.19)$$

over a region where the axion density can be treated as uniform. A realistic ALP wavefunction generally contains many modes and may be written as

$$\Psi(\mathbf{r}, t) = \sum_j C_j \exp[-i(2\pi\nu_j t - \mathbf{k}_j \cdot \mathbf{r} + \phi_j)] . \quad (1.20)$$

The coefficients C_j encode the populations of the modes and are fixed by the normalization of Ψ . The physical scalar field remains proportional to $\Re[\Psi]$ according to Equation (1.17). Moreover, if ALPs occupy many modes, the resulting field is a superposition of those modes. The interference of many modes with different velocities makes the field stochastic. The stochasticity is characterized by the co-

herence time. Chapter 2 develops this behavior for the Milky Way ALP halo and the Earth-bound ALP halo.

1.3.3 Non-gravitational interactions

An oscillating ALP field can potentially produce three non-gravitational interactions: it may couple to photons, generating an electromagnetic field; its coupling to gluons induces oscillating CP -odd nuclear moments such as a nuclear electric dipole moment (EDM); its gradient coupling to nucleons and electrons acts as oscillating magnetic fields on nuclear and electron spins [29]. All three effects oscillate at a frequency set primarily by the ALP mass and can therefore be sought with resonant precision measurements.

This thesis focuses on the gradient coupling to nucleons, also known as the “axion-wind” coupling. The nucleon Hamiltonian in the presence of an ALP field background can be written as

$$H_{\text{aNN}} = c g_{\text{aNN}} \nabla a \cdot \mathbf{I}, \quad (1.21)$$

where g_{aNN} is the coupling strength in GeV^{-1} and $\mathbf{I}$ is the nuclear spin operator. In this form of the Hamiltonian, $g_{\text{aNN}} a_0$ becomes dimensionless. The amplitude a_0 is fixed by the local dark-matter density through Equation (1.18) and has dimensions of energy. In some papers, the nuclear spin operator $\mathbf{I}$ is written as $\hbar \boldsymbol{\sigma}_n / 2$, where $\boldsymbol{\sigma}_n$ is the dimensionless Pauli spin operator.

1.4 CASPEr project

1.4.1 ALP searches with spin haloscopes

Depending on the theoretical model, different experimental schemes are adopted to search for ALP-induced spin precession [37]. Such experiments are often referred to as spin haloscopes. For example, the Global Network of Optical Magnetometers for Exotic physics searches (GNOME) [4, 36, 40, 49, 51] investigates transient exotic spin couplings. Other examples are Noble and Alkali Spin Detectors for Ultralight Coherent dark matter (NASDUCK) using a Floquet NMR quantum detector [13] and the QUaerere AXions (QUAX) experiment probing the possible ALP-electron spin interaction [20].

The Cosmic Axion Spin Precession Experiment (CASPEr) is a project proposed in 2014 [15] that uses nuclear spins to search for ALP-induced spin precession. The

project has several branches, including ALP-gluon and ALP-fermion (ALP gradient) searches, and both low-magnetic-field and high-magnetic-field implementations. The central idea is to tune the NMR resonance to a frequency that matches the ALP Compton frequency. If the two frequencies align, the ALP field applies a torque on the magnetization and induces a transverse magnetization in the sample. Since the ALP mass is unknown, the search is performed through a frequency scan: either the signal appears at a particular resonance or the absence of a signal constrains the coupling strength over the scanned band.

1.4.2 CASPER-Electric and CASPER-Comagnetometer

CASPER-Electric (CASPER-E) searches for an oscillating nuclear EDM induced by the ALP-gluon coupling. This moment generates an effective oscillating field along the nuclear spin axis in the presence of an electric field, enabling detection through NMR at the corresponding Compton frequency.

Comagnetometer configurations use two nuclear spin species in the same volume to suppress common-mode magnetic noise. The differential signal between the two species enhances sensitivity to a pseudomagnetic ALP field while rejecting electromagnetic backgrounds. Early results were obtained with a liquid-state comagnetometer [62] and an ultralow-magnetic-field configuration [28].

1.4.3 CASPER-Gradient

CASPER-Gradient or CASPER-G searches for the ALP-fermion gradient coupling described by Equation (1.21). Two setups in Mainz, Germany, cover different ALP-mass ranges. The CASPER-Gradient-LowField experiment uses bias magnetic fields up to 0.1 T, corresponding to Larmor frequencies up to 4.3 MHz. The CASPER-Gradient-Highfield experiment uses a 0-to-14.1 T magnet to reach frequencies up to 600 MHz.

A conceptual diagram of the LowField apparatus is shown in Figure 1.4. The cryostat contains a liquid helium (LHe) reservoir, excitation and pickup coils, and a SQUID magnetometer. A superconducting magnet produces the bias magnetic field B_0 , which sets the Larmor frequency and can be varied to scan ALP masses.

The resonance scheme is the main search strategy for CASPER-Gradient-LowField. The bias magnetic field is stepped through a range of values, and the NMR response is recorded at each step while no excitation is applied. A resonant signal appears as a peak in the transverse magnetization when ν_L matches ν_a .

The broadband scheme, on the other hand, continuously excites the NMR reso-

nance without changing the bias magnetic field, looking for signals that are modulated by the ALP field.

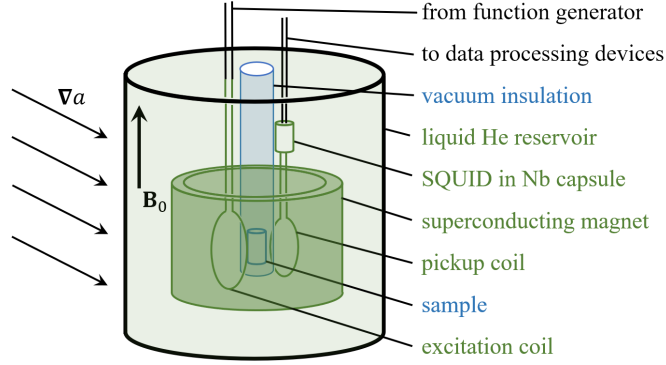


Figure 1.4: Conceptual diagram of the CASPER-Gradient-LowField setup. Superconducting devices (dark green) are submerged in a liquid helium bath (light green). The excitation and pickup coils characterize the sample and detect the spin-precession signal. The sample temperature can be controlled independently of the helium bath, enabling measurements with both thermal and hyperpolarized samples. Reproduced from Ref. [68], originally prepared by the author.

Beyond ALP-fermion coupling, CASPER-Gradient, as a sensitive magnetometer experiment, has the potential to search for other exotic physics, such as the axion-photon coupling and dark photons [11]. These possibilities will be explored in future work.

1.4.4 Candidate, exclusion, and sensitivity

There are several unknown parameters in the ALP search, including the ALP mass, the coupling strengths, and the ratio of ALP density to the local dark-matter density. In this thesis, it is assumed that ALPs constitute all of the local dark matter, and the ALP mass and coupling strengths are treated as free parameters. These two parameters define the mass-coupling parameter space.

Every haloscope is sensitive to a certain range of ALP masses. The ultimate goal for a haloscope is to detect the ALP at a particular mass and coupling strength in this parameter space. If no ALP **candidate** is found, the search sets an **exclusion** limit on the ALP coupling strength over the searched mass range. The terms “limit” and “constraint” are used interchangeably with “exclusion” in the literature. If any candidate is found, no exclusion can be set below the coupling strength at that

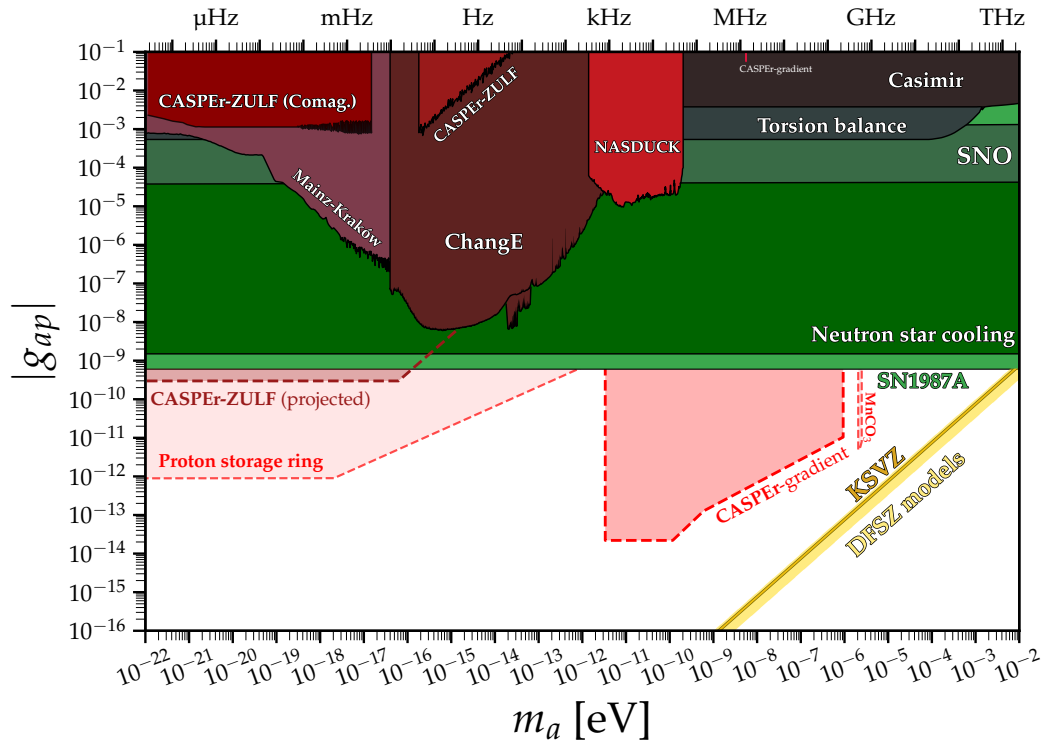


Figure 1.5: ALP-proton mass-coupling parameter space. Solid color blocks represent exclusions. Projections of sensitivities are indicated by the dashed lines on the edges of the blocks. Reproduced from Ref. [45] with permission.

mass. The common practice is to leave gaps or openings in the exclusion to indicate the presence of candidates.

Besides exclusion, a haloscope can also report its **sensitivity** in the mass-coupling parameter space. The sensitivity in the parameter space is determined by the mass range that the haloscope can cover and the coupling strength that can be detected at each mass. Even before any search is performed, the **projected sensitivity** can be calculated based on the experimental parameters and the expected ALP signal.

Figure 1.5, as an example of the parameter space searched by haloscopes, summarizes the current status of the ALP-proton mass-coupling parameter space.

The material in this chapter concerning the Milky Way halo model is adapted in part from Ref. [68], a publication by the author and collaborators.

This chapter discusses two models of ALP dark matter: the Milky Way halo model and the Earth-bound halo model. The Milky Way halo model is based on the standard halo model (SHM) of dark matter, which assumes that the dark matter in the Milky Way is virialized and follows a Maxwell–Boltzmann velocity distribution. The Earth-bound halo model considers the possibility of a local overdensity of ALP dark matter bound to Earth, which could produce experimental signatures different from those predicted by the Milky Way halo model.

Analyzing these models provides insight into the signal signatures expected in experiments searching for ALP dark matter.

2.1 Milky Way halo

2.1.1 Virialized dark matter halo

In the SHM [27], the Milky Way dark matter population is treated as a virialized² ensemble. In the Galactic rest frame, the dark-matter velocity distribution is approximated by a Maxwell–Boltzmann distribution truncated near the Galactic escape speed [27]. The average speed of local dark matter³ is $v_{\text{DM}} \approx 220 \text{ km s}^{-1}$. The local dark matter density is estimated to be $\rho_{\text{DM}}^{\text{MW}} \approx 0.4 \text{ GeV cm}^{-3}$. The Rabi frequency Ω_a can be computed from Equations (1.18), (1.19), and (1.21):

$$\Omega_a = \frac{1}{2\hbar} g_{a\text{NN}} a_0 m_a c v_{\perp} \approx \frac{1}{\sqrt{2}} g_{a\text{NN}} \sqrt{\hbar c \rho_{\text{DM}}^{\text{MW}}} v_{\perp} \quad (2.1)$$

$$= 2\pi \times 2 \times 10^{-10} \text{ Hz} \frac{g_{a\text{NN}}}{10^{-9} \text{ GeV}^{-1}} \left(\frac{\rho_{\text{DM}}^{\text{MW}}}{0.4 \text{ GeV cm}^{-3}} \right)^{1/2} \frac{v_{\perp}}{220 \text{ km s}^{-1}}. \quad (2.2)$$

Here it is assumed that the ALP-field gradient comes exclusively from the relative motion of the field and the detector. It is important to note that the ALP field is a stochastically varying quantity [18], so that the Ω_a in Equation (2.2) is the root-mean-square (rms) value. For $v_{\perp} = 220 \text{ km s}^{-1}$ and $g_{a\text{NN}} \leq 10^{-9} \text{ GeV}^{-1}$, the Rabi period is $2\pi/\Omega_a \geq 100 \text{ yr}$, much longer than any practical measurement duration; the ALP-field gradient is therefore a weak drive on the spins.

The ALP gradient coupling can also be interpreted as an effective oscillating pseudomagnetic field B_a acting on the nuclear spins. The amplitude of the pseudomagnetic field can be estimated from the Rabi frequency Ω_a and the gyromagnetic ratio γ of the nuclear spin:

$$B_a = \frac{\Omega_a}{\gamma} \quad (2.3)$$

$$= 2 \times 10^{-17} \text{ T} \frac{\gamma}{-2\pi \times 12 \text{ MHz T}^{-1}} \frac{g_{a\text{NN}}}{10^{-9} \text{ GeV}} \times \left(\frac{\rho_{\text{DM}}^{\text{MW}}}{0.4 \text{ GeV cm}^{-3}} \right)^{1/2} \frac{v_{\perp}}{220 \text{ km s}^{-1}}. \quad (2.4)$$

Here $(-2\pi) \times 12 \text{ MHz T}^{-1}$ is the gyromagnetic ratio of the nuclear spin of ^{129}Xe .

² The term “virialized” means that the dark matter is gravitationally bound.

³ Here, “local” refers to the solar radius in the Milky Way.

In this way of thinking, the ALP gradient coupling is equivalent to a very weak oscillating magnetic field.

Although it is called the “standard” halo model, it is not a complete description of the Milky Way dark matter halo, but focuses on the dark matter properties at the solar radius. It is more of a model built on phenomenological observations, rather than a model tested through direct measurements. Not every question about the Milky Way halo can be answered by the SHM. For example, the SHM does not account for the presence of substructure in galaxies. There have been alternative models or modifications to the SHM, including SHM⁺⁺ [27], caustic rings [54], and the presence of substructure in the form of streams [46]. In these models, the local dark matter density and velocity distribution can be significantly different from those in the SHM. These alternative models will be accounted for in future CASPER work.

2.1.2 Stochasticity due to interference

If the ALP constitutes the dominant component of the Milky Way dark matter, the local ALP field is a superposition of many modes with different velocities and phases. Here the phases of the occupied modes are assumed to be uncorrelated. Each velocity class contributes one term to the multimode field in Equation (1.20), with

$$v(\mathbf{v}) = \frac{m_a c^2}{h} \left(1 + \frac{v^2}{2c^2} \right), \quad k(\mathbf{v}) = \frac{m_a \mathbf{v}}{\hbar}. \quad (2.5)$$

The relative spread of velocities is on the order of $v_a/c \sim 10^{-3}$. Due to the second-order Doppler effect, the relative spread of kinetic energies is on the order of $(v_a/c)^2 \sim 10^{-6}$. The inverse of this spread is defined as the quality factor of the ALP field, $Q_a = (c/v_a)^2 \sim 10^6$. Therefore, in the frequency domain, the ALP field is a narrowband signal with a fractional linewidth $\Delta v_a/v_a = Q_a^{-1} \sim 10^{-6}$. Considering the Maxwell-Boltzmann velocity distribution, the sum of the modes shown in Equation (1.20) is a random walk in the complex Ψ plane: the two quadratures approach Gaussian random variables, while the field (average) envelope follows a Rayleigh distribution [18, 31]. Consequently, in the time domain, the ALP field is a coherent oscillation only on times short compared with its coherence time; on longer time scales, the amplitude and phase drift stochastically, due to the spread of frequencies in the multimode superposition. The ALP coherence time τ_a can be estimated from the quality factor or the ALP linewidth:

$$\tau_a = \frac{Q_a}{\pi v_a} = \frac{1}{\pi \Delta v_a}. \quad (2.6)$$

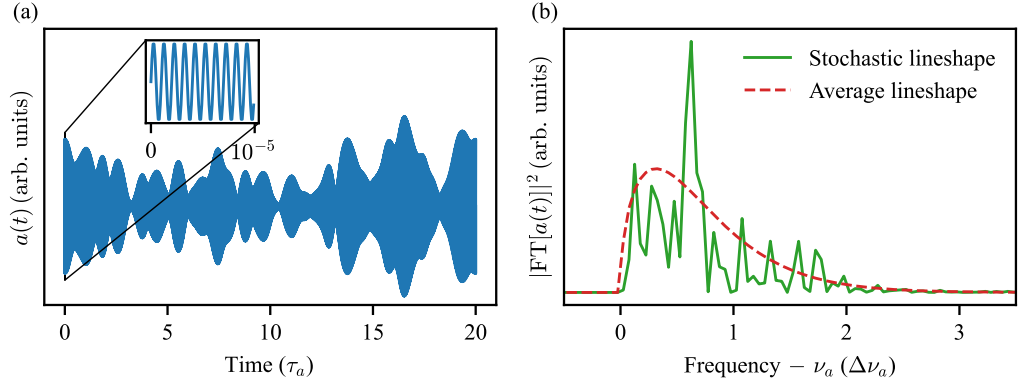


Figure 2.1: Simulated axion fields in the time and frequency domains. (a) Time-domain axion field $a(t)$, with a magnified view showing the oscillation at the axion Compton frequency in the inset. The envelope of $a(t)$ changes over time, indicating that the coherence time τ_a is finite. Here τ_a is chosen as $1/(\pi\Delta\nu_a)$. (b) The stochastic lineshape in the frequency domain is computed as $|\text{FT}[a(t)]|^2$, where $\text{FT}[a(t)]$ denotes the Fourier transform of the time-domain signal shown in (a). The average lineshape represents the expected amplitude spectrum after sufficient averaging. The linewidth $\Delta\nu_a$ is approximated as ν_a/Q_a . Reproduced from Ref. [7], originally prepared by the author.

An example of a stochastic ALP field in the time and frequency domains is shown in Figure 2.1.

2.1.3 Periodic modulations

In the resonance scheme, CASPER-Gradient is only sensitive to the component of ∇a perpendicular to the bias magnetic field B_0 .⁴ The bias magnetic field is always vertical with respect to the ground. Therefore, the ALP-gradient-driven spin precession depends on the direction of the relative velocity between the Earth and the local dark-matter halo. The Earth’s rotation and orbital motion modulate this relative velocity and give rise to daily and annual modulations of the ALP signal, respectively.

As the Earth rotates, this direction sweeps through space with a period of one sidereal day. An ALP signal therefore exhibits a daily modulation of its amplitude and phase [31]. In addition to daily rotation, the Earth’s orbital velocity around the Sun modulates the relative ALP-detector velocity with a period of one year. Both

⁴ This is sometimes defined as the sensitive axis of a haloscope. However, the term “sensitive axis” is also defined as the axes perpendicular to the bias magnetic field. Due to the ambiguity in the definition, the term “sensitive axis” is avoided in this thesis.

the daily and annual modulations have precisely predictable periods and phases, making them useful consistency checks for any claimed dark-matter-candidate signal. Examples of daily and annual modulations are shown in Figure 2.2.

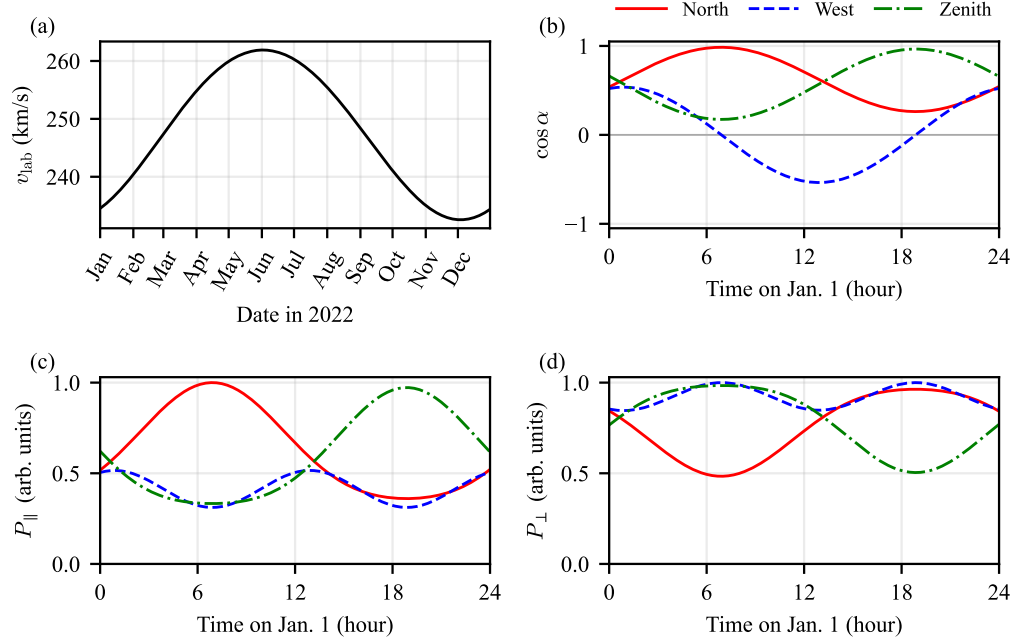


Figure 2.2: Periodic (annual and daily) modulations. (a) Annual modulation of laboratory speed due to the Earth’s orbital motion around the Sun. With the Sun’s peculiar velocity added as separate velocity components, the maximum speed of the laboratory in the Galactic rest frame is approximately 260 km s^{-1} , larger than previously calculated values. (b) Daily modulations of angles between the ALP field and the three orthogonal orientations: towards the north, the west, and the zenith. The location is Boston University and the date is January 1 (UTC time). (c) Daily modulations of $P_{\parallel}$, the power of the ALP field gradient parallel to the three orientations with regard to an external magnetic field. The power is normalized such that the maximum value is 1 (the scaling factor is the same for the three cases). (d) Similar to panel (c) but for the ALP field gradient perpendicular to the three orientations. The plot is reproduced from Ref. [31] with permission.

2.2 Earth-bound ALPs

A distinct source arises from ALPs that are gravitationally bound to the Earth [8]. Such Earth-bound states are analogous to atomic bound states—the Earth’s gravitational potential plays the role of the Coulomb well and the ALP plays the role of the electron—and can support a discrete spectrum of spatial eigenstates. If additional ALPs can be trapped by the Earth, they could produce a local overdensity relative to the galactic dark matter density. This section considers the overdensity, derives the properties of these bound states, estimates the resulting spin-precession signal, and discusses the characteristic daily modulation that distinguishes this signal from the galactic background.

2.2.1 Trapping and accumulation of dark matter

It is assumed here that the dark matter was produced in the early universe and is now gravitationally bound to the Milky Way. ALPs are assumed to constitute most of the dark matter in the galactic halo. This thesis considers the possibility that a fraction of the galactic ALPs can be trapped by the Earth’s gravitational potential, forming a local overdensity. If the SHM is assumed, the local galactic dark matter density is $\rho_{\text{DM}}^{\text{MW}} \sim 0.4 \text{ GeV/cm}^3$, and the typical velocity and velocity spread of dark matter are $\sim 220 \text{ km/s}$. Such velocities are much larger than the escape velocity of the Earth, 11.2 km/s . Therefore, ALPs in the galactic halo must cool down (lose enough kinetic energy) so that they can be captured by the Earth’s gravitational potential.

The cooling mechanism determines how many ALPs can be captured and how the ALP population is distributed in the bound states. Dark matter is usually assumed to be collisionless, so that gravitational interactions are consequently assumed inefficient in cooling ALPs. Possible cooling mechanisms involving non-gravitational interactions include dark matter self-interactions. These are also referred to as “friction”, since they can dissipate the kinetic energy of the ALP. There are also theories suggesting that gravitational interactions alone may be sufficient to trap ALPs. For example, gravitational interactions between ultralight particles such as ALPs can lead to the rapid formation of compact objects, such as supermassive black holes [55]. It may be reasonable to expect gravitational interactions to also lead to the formation of an Earth-bound ALP halo.

This thesis does not focus on the mechanism of trapping and accumulation of ALPs, but rather assumes that a fraction of ALPs can be trapped by the Earth, and focuses on the properties of the bound states and the resulting spin-precession signal.

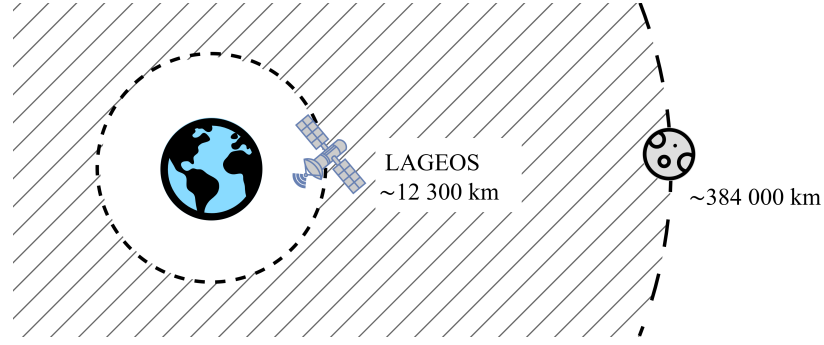


Figure 2.3: Illustration of the measurements in Ref. [3].

The total mass of dark matter trapped in Earth-bound states is constrained by astrophysical and geophysical observations. Reference [3] used lunar laser-ranging and the motion of the LAGEOS satellite to determine the mass between the LAGEOS satellite orbit and the Moon's orbit. The orbits of the Moon and the LAGEOS satellite and the dark matter measured are illustrated in Figure 2.3. The measured mass of dark matter between the LAGEOS satellite orbit and the Moon's orbit is

$$GM_{\text{DM}} = (0.3 \pm 4) \times 10^{-9} GM_{\oplus}, \quad (2.7)$$

where $M_{\oplus}$ is the mass of Earth. The uncertainty in Equation (2.7), $4 \times 10^{-9} M_{\oplus}$, corresponds to an upper limit on the mass between the orbits of the Moon and the LAGEOS satellite. If there is indeed mass there, it is likely to be dark matter, since the mass of ordinary matter in this region is negligible. Though the dark matter mass inside the LAGEOS satellite orbit is not directly constrained by this measurement, it is reasonable to neglect it so that the upper limit of dark matter mass enclosed by the Moon's orbit is not overestimated.

2.2.2 Gravitational potential and the quantum picture

To simplify the calculation, Earth is approximated as a spherical object in this thesis. The Earth's density varies with radius. From geophysical measurements of seismic waves, the Preliminary Reference Earth Model (PREM) [25] was constructed, giving the Earth's density profile as shown in Figure 2.4. The Earth's escape velocity can therefore be found by

$$v_{\text{escape}} = \sqrt{\frac{2GM_{\oplus}}{R_{\oplus}}}, \quad (2.8)$$

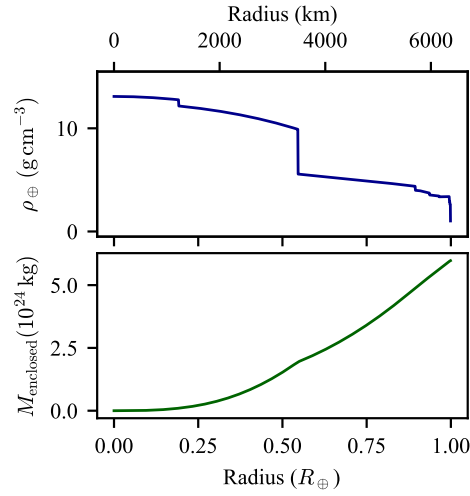


Figure 2.4: Earth’s density (upper plot) and enclosed mass (lower plot) as functions of radius according to PREM [25].

where $M_{\oplus} = 6 \times 10^{24}$ kg and $R_{\oplus} = 6378.1$ km are the mass and average radius of Earth, respectively. By substituting these values, the Earth’s escape velocity is found to be 11.2 km/s. Since the escape speed is much smaller than the speed of light, the nonrelativistic limit is assumed in this discussion.

The gravitational potential of Earth, taking the potential at infinity to be zero, is obtained from the spherically symmetric density profile:

$$V_g(r) = -Gm \left[\frac{M_{\text{enclosed}}(r)}{r} + 4\pi \int_r^{\infty} \rho_{\oplus}(r') r' dr' \right] \quad (2.9)$$

where r is the distance measured from the Earth’s center, G is Newton’s constant, $M_{\text{enclosed}}(r)$ is the Earth’s mass enclosed within radius r , m is the mass of the object experiencing the gravitational field, and $\rho_{\oplus}(r)$ is the Earth’s density profile. For $r \geq R_{\oplus}$, this expression reduces to $V_g(r) = -GM_{\oplus}m/r$; inside the Earth, the integral accounts for the contribution from the outer shells. The potential as a function of radius can be calculated numerically based on PREM, as shown in Figure 2.5.

Ultralight dark matter particles bound to Earth can no longer be treated as classical objects because of their large de Broglie wavelengths. A rough estimate of the wavelength is made as follows. The ALP masses that CASPER-Gradient can probe range from ~ 4 peV to 2.5 μ eV. Consider ALPs trapped by the Earth’s potential with velocities roughly below the escape velocity. The de Broglie wavelength as a function of particle mass $\lambda_{\text{dB}}(m_a, v_a) = h/(m_a v_a)$ gives [66]

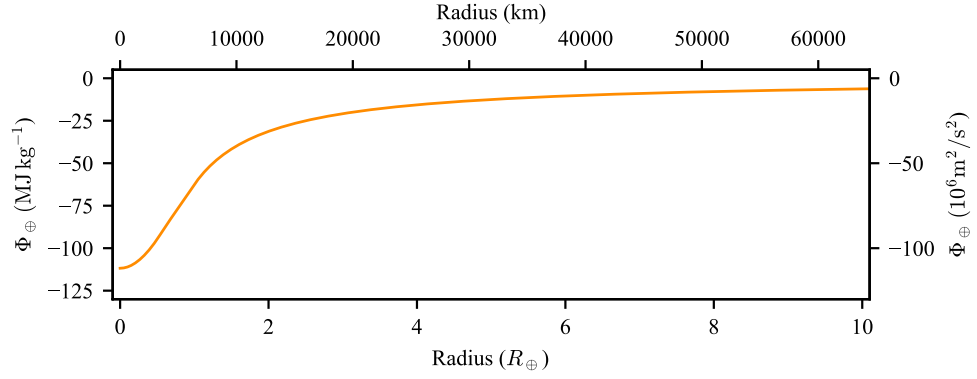


Figure 2.5: Earth's gravitational potential as a function of radius, accounting for the density profile.

- $\lambda_{\text{dB}}(4 \text{ peV}, 11.2 \text{ km/s}) \approx 8.3 \times 10^9 \text{ m} \approx 1300 R_{\oplus}$ and
- $\lambda_{\text{dB}}(2.5 \text{ } \mu\text{eV}, 11.2 \text{ km/s}) \approx 13 \text{ km} \approx 0.0021 R_{\oplus}$.

Therefore, across the scan range of CASPER, the de Broglie wavelength spans values from much smaller to much larger than the Earth's radius. In later chapters, a measurement probing $\sim 1 \text{ MHz}$ ALPs is presented. This corresponds to a de Broglie wavelength $\sim 1 R_{\oplus}$ [66]. To prepare for the analysis, this scenario is considered in the following sections, and the quantum picture of ALPs bound to Earth is discussed.

2.2.3 Bound-state wavefunctions

The general relation between an ALP wavefunction, its local energy density, and the real scalar field was established in Equations (1.16) and (1.17). For the Earth-bound model, the wavefunction is expanded in the discrete gravitational eigenstates:

$$\Psi(\mathbf{r}, t) = e^{-im_a c^2 t/\hbar} \sum_{nlm} c_{nlm} e^{-iE_n t/\hbar} \psi_{nlm}(\mathbf{r}). \quad (2.10)$$

Here $\Psi(\mathbf{r}, t)$ is the time-dependent wavefunction, ψ_{nlm} and E_n are the eigenfunction and energy of the state (n, l, m) , and c_{nlm} are expansion coefficients determined by the state populations. The total particle number N_a enters through the density below. The fast oscillation at the ALP Compton frequency is indicated by $e^{-im_a c^2 t/\hbar}$, while the small shift in oscillation frequency is indicated by $e^{-iE_n t/\hbar}$. The corresponding local energy density, $\rho_{DM}(\mathbf{r}) = m_a c^2 N_a \langle |\Psi(\mathbf{r}, t)|^2 \rangle_t$, fixes the field

amplitude through Equation (1.17). The remaining task is therefore to determine the eigenfunctions, eigenenergies, and state populations in the Earth's potential.

In the non-relativistic limit, the eigenstates can be found by solving the time-independent Schrödinger equation (TISE)

$$\left[-\frac{\hbar^2}{2m_a} \nabla^2 + V_g(r) \right] \psi_{nlm}(r) = E_n \psi_{nlm}(r). \quad (2.11)$$

At infinity ($r \rightarrow \infty$), the potential vanishes. In analogy to the hydrogen (proton-electron) atom, the ALP-Earth system is a “gravitational atom”.

Considering the spherical symmetry of the Earth's gravitational potential, a spherical coordinate system (r, θ, φ) is adopted, where r is the radial distance from the origin, θ is the polar angle, and φ is the azimuthal angle. In spherical coordinates,

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \varphi^2}, \quad (2.12)$$

so that the TISE can be rewritten as

$$\left[-\frac{\hbar^2}{2m} \left(\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) - \frac{\hat{L}^2}{\hbar^2 r^2} \right) + V_g(r) \right] \psi_{nlm}(r, \theta, \varphi) = E_n \psi_{nlm}(r, \theta, \varphi), \quad (2.13)$$

where $\hat{L}$ is the angular momentum operator. The eigenfunction can be separated as

$$\psi_{nlm}(r) = R_{nl}(r) Y_l^m(\hat{r}), \quad (2.14)$$

where R_{nl} is the radial wavefunction and Y_l^m is the spherical harmonic. The spherical harmonics satisfy

$$\hat{L}^2 Y(\theta, \varphi) = l(l+1) \hbar^2 Y(\theta, \varphi). \quad (2.15)$$

Therefore, to derive the radial wavefunction of an eigenstate with angular quantum number l , the TISE is transformed to

$$-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \frac{\partial R_{nl}(r)}{\partial r} \right] + \left[\frac{l(l+1) \hbar^2}{2mr^2} + V_g \right] R_{nl}(r) = E_n R_{nl}(r). \quad (2.16)$$

Since the Earth's density is not uniform, it is difficult to derive the radial wavefunction analytically. Hence, a numerical method is adopted. In a discretized space, a matrix for the Hamiltonian in Equation (2.16) is constructed. The eigenval-

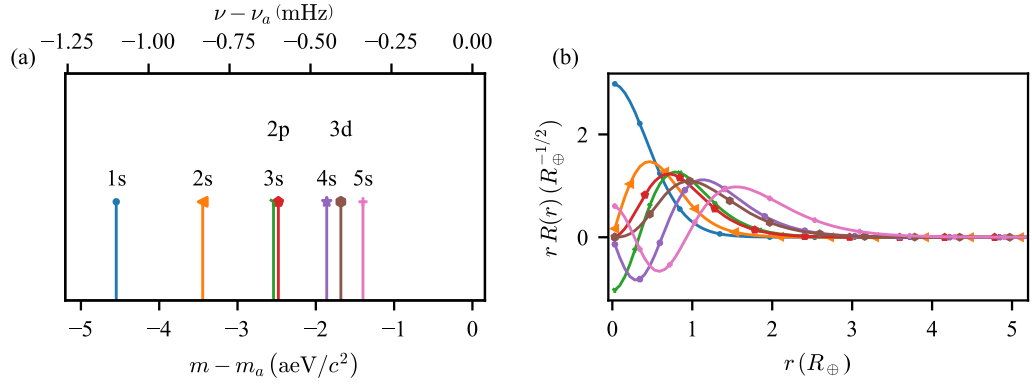


Figure 2.6: The lowest 7 eigenstates derived from the TISE assuming an 1 MHz axion Compton frequency. (a) Eigenenergies converted to masses and frequencies compared to the axion rest mass and Compton frequency. (b) The corresponding reduced radial wavefunctions (radius times the radial wavefunction) using the same color and marker scheme as in (a).

ues (eigenenergies) and eigenvectors (eigenstates) can be derived using the Python package SciPy. This method is implemented in the Python class `EarthBoundAxionHalo` of `axionbloch` [64].

Note that normalization is necessary for deriving the correct magnitude of the eigenfunctions. The wavefunctions are normalized so that the total probability integrates to unity,

$$\int |\psi_{nlm}(r)|^2 d^3r = \int_0^\infty |R_{nl}(r)|^2 r^2 dr \int |Y_l^m(\hat{r})|^2 d\Omega = 1. \quad (2.17)$$

Here $d\Omega$ denotes the solid-angle element. Examples of normalized wavefunctions are illustrated in Figure 2.6.

2.2.4 ALP field gradient

The ALP field gradient at the detector can be derived from the spatial gradient of the wavefunction ψ_{nlm} evaluated at the detector position. The gradient operator in spherical coordinates is

$$\nabla = \hat{r} \frac{\partial}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\phi} \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}. \quad (2.18)$$

The eigenstates are defined in the Earth-rest frame, whereas the gradient coupling is measured in the laboratory frame. The ALP field is a Lorentz scalar⁵, while its derivatives transform according to the inverse Lorentz transformation. The chain rule, $\partial'_\mu = (\Lambda^{-1})^\nu_\mu \partial_\nu$, mixes the temporal derivative with the component of the spatial gradient parallel to the laboratory velocity. Here Λ^μ_ν is the 4×4 Lorentz-boost matrix relating the Earth-rest-frame coordinates $x^\mu = (ct, \mathbf{r})$ to the laboratory-frame coordinates through $x'^\mu = \Lambda^\mu_\nu x^\nu$. Its inverse appears because the derivative basis transforms oppositely to the coordinates; the Greek indices run over the temporal and three spatial components. To obtain the spatial transformation explicitly, the unboosted gradient is decomposed into components parallel and perpendicular to the velocity of the laboratory in the Earth-rest frame, $\mathbf{v}$:

$$\nabla_\parallel a = \frac{\mathbf{v}}{v^2}(\mathbf{v} \cdot \nabla a), \quad \nabla_\perp a = \nabla a - \nabla_\parallel a. \quad (2.19)$$

The inverse Lorentz boost leaves the perpendicular component unchanged and transforms the parallel component as

$$(\nabla_{\text{lab}} a)_\perp = \nabla_\perp a, \quad (\nabla_{\text{lab}} a)_\parallel = \gamma_{\text{lab}} \left(\nabla_\parallel a + \frac{\mathbf{v}}{c^2} \partial_t a \right), \quad (2.20)$$

where $\nabla_{\text{lab}} a$ is the ALP gradient in the laboratory frame, and $\gamma_{\text{lab}} = (1 - v^2/c^2)^{-1/2}$. Adding these two components gives

$$\nabla_{\text{lab}} a = \nabla a + \frac{\gamma_{\text{lab}} - 1}{v^2} \mathbf{v}(\mathbf{v} \cdot \nabla a) + \gamma_{\text{lab}} \frac{\mathbf{v}}{c^2} \partial_t a. \quad (2.21)$$

In the limit $v/c \rightarrow 0$, the laboratory and Earth-rest frames coincide. Although the second term in Equation (2.21) contains v^2 in the denominator, $v/c \rightarrow 0$ would not make the term diverge because when $v/c \rightarrow 0$, $(\gamma_{\text{lab}} - 1) \propto v^2/c^2$, and the complete term has the regular limit

$$\lim_{v \rightarrow 0} \frac{\gamma_{\text{lab}} - 1}{v^2} \mathbf{v}(\mathbf{v} \cdot \nabla a) = 0. \quad (2.22)$$

The temporal-gradient term also vanishes in the limit $v/c \rightarrow 0$, so Equation (2.21) reduces to

$$\nabla_{\text{lab}} a = \nabla a, \quad (2.23)$$

as expected.

5 A Lorentz scalar is a physical quantity whose value is unchanged under Lorentz transformations between inertial observers.

For $0 < v/c \ll 1$, the Lorentz factor can be expanded as

$$\gamma_{\text{lab}} = 1 + \frac{1}{2} \frac{v^2}{c^2} + O\left(\frac{v^4}{c^4}\right), \quad \frac{\gamma_{\text{lab}} - 1}{v^2} = \frac{1}{2c^2} + O\left(\frac{v^2}{c^4}\right). \quad (2.24)$$

The condition $v/c \ll 1$ does not necessarily make the temporal-gradient contribution negligible because $\partial_t a$ contains the rapid ALP Compton oscillation. It is the laboratory-frame gradient $\nabla_{\text{lab}} a$ that enters the interaction Hamiltonian in Equation (1.21). The unboosted gradient can be inferred from Equations (1.16) and (2.10) as

$$\nabla a(\mathbf{r}, t) = \sqrt{\frac{N_a \hbar^3 c}{2m_a}} \Re[\nabla \Psi(\mathbf{r}, t)] \quad (2.25)$$

$$= \sqrt{\frac{N_a \hbar^3 c}{2m_a}} \Re \left[e^{-\frac{im_a c^2 t}{\hbar}} \sum_{nlm} c_{nlm} e^{-\frac{iE_n t}{\hbar}} \nabla \psi_{nlm}(\mathbf{r}) \right], \quad (2.26)$$

where

$$\nabla \psi_{nlm}(\mathbf{r}) = \hat{\mathbf{r}} \frac{\partial R_{nl}}{\partial r} Y_l^m + \hat{\boldsymbol{\theta}} \frac{R_{nl}}{r} \frac{\partial Y_l^m}{\partial \theta} + \hat{\boldsymbol{\phi}} \frac{R_{nl}}{r \sin \theta} \frac{\partial Y_l^m}{\partial \varphi}. \quad (2.27)$$

Using Equation (2.10), the two Lorentz-boost terms in Equation (2.21) become

$$\frac{\gamma_{\text{lab}} - 1}{v^2} \mathbf{v}(\mathbf{v} \cdot \nabla a) = \sqrt{\frac{N_a \hbar^3 c}{2m_a}} \Re \left[e^{-\frac{im_a c^2 t}{\hbar}} \sum_{nlm} c_{nlm} e^{-\frac{iE_n t}{\hbar}} \frac{\gamma_{\text{lab}} - 1}{v^2} \mathbf{v}(\mathbf{v} \cdot \nabla \psi_{nlm}) \right], \quad (2.28)$$

and

$$\gamma_{\text{lab}} \frac{\mathbf{v}}{c^2} \partial_t a = \sqrt{\frac{N_a \hbar^3 c}{2m_a}} \Re \left[e^{-\frac{im_a c^2 t}{\hbar}} \sum_{nlm} c_{nlm} e^{-\frac{iE_n t}{\hbar}} \left(-i\gamma_{\text{lab}} \frac{m_a c^2 + E_n}{\hbar c^2} \mathbf{v} \psi_{nlm} \right) \right]. \quad (2.29)$$

The size of these corrections can be estimated from the rotation of the Earth. The Earth's radius $R_{\oplus} \simeq 6.37 \times 10^6$ m and the sidereal rotation period $T_{\oplus} \simeq 8.62 \times 10^4$ s give the maximum surface speed [66]

$$v_{\text{rot}}^{\text{max}} = \frac{2\pi R_{\oplus}}{T_{\oplus}} \simeq 465 \text{ m s}^{-1} = 1.55 \times 10^{-6} c. \quad (2.30)$$

The corresponding Lorentz factor satisfies

$$\gamma_{\text{lab}} - 1 \simeq \frac{1}{2} \left(\frac{v_{\text{rot}}^{\text{max}}}{c} \right)^2 \simeq 10^{-12}, \quad (2.31)$$

so the correction in Equation (2.28) is [66]

$$\left| \frac{\gamma_{\text{lab}} - 1}{v^2} \mathbf{v}(\mathbf{v} \cdot \nabla a) \right| \approx 10^{-12} \left(\frac{v}{v_{\text{rot}}^{\text{max}}} \right) |\nabla a|. \quad (2.32)$$

Concerning the other correction to the ALP gradient in Equation (2.29), using Equation (2.26) and $|\nabla \psi_{nlm}| \approx |\psi_{nlm}|/(R_{\oplus})$ at $\nu_a = 1$ MHz, the correction is estimated as [66]

$$\left| \gamma_{\text{lab}} \frac{\mathbf{v}}{c^2} \partial_t a \right| \approx \frac{2\pi}{c^2} \gamma_{\text{lab}} \nu_a v R_{\oplus} |\nabla a| \approx 10^{-1} \left(\frac{v}{v_{\text{rot}}^{\text{max}}} \right) |\nabla a|, \quad (2.33)$$

which is more significant than the first correction.

The laboratory velocity relative to the Earth-bound halo is written generally as

$$\mathbf{v} = v_r \hat{\mathbf{r}} + v_{\theta} \hat{\boldsymbol{\theta}} + v_{\varphi} \hat{\boldsymbol{\varphi}}, \quad v^2 = v_r^2 + v_{\theta}^2 + v_{\varphi}^2, \quad (2.34)$$

where the radial, polar, and azimuthal components are all retained. The boost correction to the complex gradient amplitude of one mode is then

$$\begin{aligned} \delta \mathcal{G}_{nlm} = & \frac{\gamma_{\text{lab}} - 1}{v^2} \mathbf{v} \left(v_r \frac{\partial R_{nl}}{\partial r} Y_l^m + v_{\theta} \frac{R_{nl}}{r} \frac{\partial Y_l^m}{\partial \theta} + v_{\varphi} \frac{R_{nl}}{r \sin \theta} \frac{\partial Y_l^m}{\partial \varphi} \right) \\ & - i \gamma_{\text{lab}} \frac{m_a c^2 + E_n}{\hbar c^2} \mathbf{v} R_{nl} Y_l^m. \end{aligned} \quad (2.35)$$

For the spherically symmetric states with $l = m = 0$, the angular derivatives vanish, but the first correction retains a radial contribution proportional to $v_r \partial R_{n0}/\partial r$. It vanishes only when $v_r = 0$, while the contribution from the temporal derivative remains. For other states with $l > 0$, the angular derivatives are generally non-zero, so the first correction may have contributions from polar and azimuthal components of the laboratory velocity, and the ALP gradient varies with the detector location and time since the wavefunction is not spherically symmetric and Earth is moving. If multiple modes are populated, the total gradient is a superposition of the gradients from each mode, with weights corresponding to the population distribution, as can be seen in Equation (2.26).

A rough estimate of the Rabi frequency $\Omega_a^{\oplus}$ of the pseudomagnetic field can

be made using Equation (2.2) and the escape speed v_{esc} as the speed of the Earth-bound ALP [66]

$$\Omega_a^\oplus \approx 2\pi \times 0.6 \text{ mHz} \frac{g_{\text{aNN}}}{10^{-9} \text{ GeV}^{-1}} \left(\frac{M_{\text{DM}}^{\text{EH}}}{4 \times 10^{-9} M_\oplus} \right)^{1/2} \frac{v}{11.2 \text{ km s}^{-1}}, \quad (2.36)$$

where $M_{\text{DM}}^{\text{EH}}$ is the total mass of the Earth-bound ALP, and v is the speed of the ALP at the detector location. A more accurate estimate can be made using the numerical values of the wavefunction gradients at the detector location. With Equations (2.4, 2.26, 2.27), it can be estimated as [66]

$$\Omega_a^\oplus = c g_{\text{aNN}} \sqrt{\frac{N_a \hbar^3 c}{2m_a}} |(\nabla \psi_{nlm})_\perp|, \quad (2.37)$$

where $|(\nabla \psi_{nlm})_\perp|$ is the magnitude of the wavefunction gradient component perpendicular to the bias magnetic field B_0 [i.e., the $\hat{\theta}$ and $\hat{\phi}$ terms from Equation (2.27)]. Using the numerical values of the wavefunction gradients, the value can be obtained as

$$\Omega_a^\oplus \approx 2\pi \times 1 \text{ mHz} \frac{g_{\text{aNN}}}{10^{-9} \text{ GeV}^{-1}} \left(\frac{M_{\text{DM}}^{\text{EH}}}{4 \times 10^{-9} M_\oplus} \right)^{1/2}. \quad (2.38)$$

The two estimates in Equations (2.36) and (2.38) yield consistent results within an order of magnitude. The precise values of the gradients can be computed numerically from the eigenstates and are used to estimate the expected signal strength in Chapter 4. An example of ALP gradients and Rabi frequencies for the $|n = 2, l = 1, m = 0\rangle$ state is illustrated in Figure 2.7. As can be seen from the figure, the radial component of the Rabi frequency is the largest due to the strong radial gradient of the Earth's gravitational potential. This Rabi frequency is ~ 7 orders of magnitude larger than the expected Rabi frequency from the galactic halo ALP, as estimated after Equation (2.2). If such an amount of axionlike dark matter can be accumulated in the Earth's gravitational potential, it would produce a much stronger signal than the galactic halo ALP, making it a target that CASPER-Gradient is sensitive to.

2.2.5 Population, interference and coherence

Unlike the galactic halo signal, which has a continuous Maxwell–Boltzmann spectrum, Earth-bound ALPs occupy discrete energy levels. Only the lowest few eigenstates are of interest because the higher eigenstates have small wavefunction am-

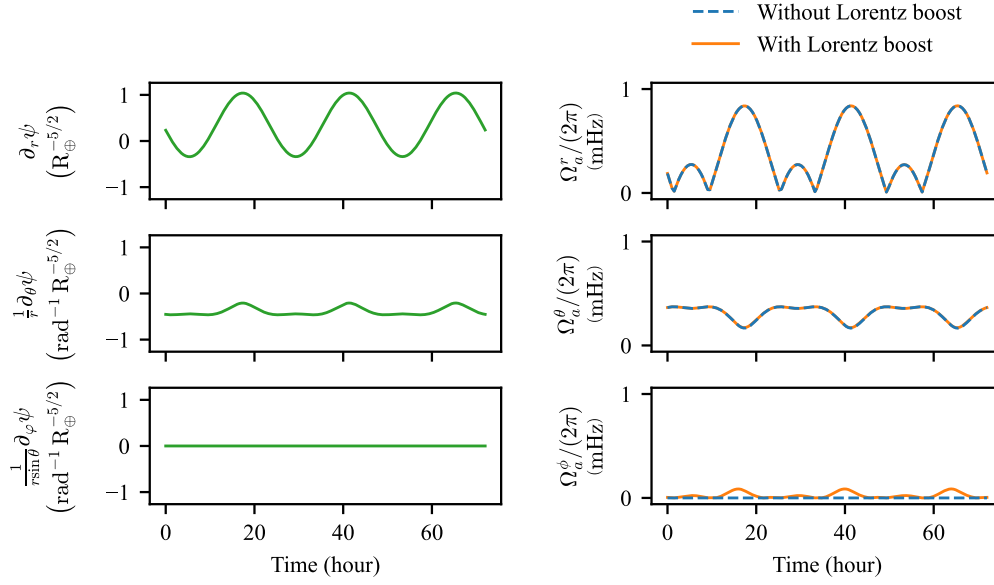


Figure 2.7: Daily modulation of the gradient component for the $|n = 2, l = 1, m = 0\rangle$ state at the location of CASPEr-Gradient (Helmholtz Institute Mainz) from 07:00 (Central European Time), December 13, 2022. The modulation is due to the rotation of the Earth. The quantization axis is arbitrarily chosen along the Earth–Sun line. Calculations with and without a Lorentz boost are shown for comparison. In general, the Lorentz boost modifies the Rabi frequency by less than 1 mHz at an ALP Compton frequency of 1 MHz. Note that Ω_a^ϕ is zero without a Lorentz boost (since the magnetic quantum number is $m = 0$) but becomes non-zero with the boost due to the Earth’s rotation.

plitudes at the detector location. The population of the eigenstates is determined by the dark matter capture and accumulation processes and potential ALP self-interaction or interactions with other particles, which are under investigation by the CASPER collaboration. The population distribution is not explored in this thesis. Empirical results are presented in Chapter 4 for the scenario where most ALPs are in the lowest eigenstates.

Each level n contributes an ALP oscillation at a slightly different Compton frequency

$$\nu_n = \nu_a + \frac{|E_n|}{h}, \quad (2.39)$$

where $\nu_a = m_a c^2 / h$ is the rest-mass Compton frequency. If multiple modes are populated simultaneously, the phases of the individual oscillations drift relative to each other over time. This leads to a finite coherence time for the multimode ALP field. A rough estimate of the coherence time can be made from the spacing ΔE between eigenstates:

$$\tau_{\text{a, est}} \approx \frac{1}{\Delta \nu} = \frac{h}{\Delta E}. \quad (2.40)$$

The maximum spacing can be estimated from the ALP mass and the Earth's gravitational potential which is $|\Delta V_g| = 125 \text{ MJ kg}^{-1}$ [66]:

$$\Delta E \approx m_a \Delta V_g \approx m_a \frac{GM_\oplus}{R_\oplus} \quad (2.41)$$

$$= 6 \times 10^{-18} \text{ eV} \frac{\nu_a}{1 \text{ MHz}}. \quad (2.42)$$

The corresponding frequency spacing and coherence time are

$$\Delta \nu \approx \frac{\Delta E_{\text{max}}}{h} \approx 10^{-9} \nu_a \quad (2.43)$$

and

$$\tau_{\text{a, est}} \approx \frac{1}{\Delta \nu} = \frac{10^9}{\nu_a}. \quad (2.44)$$

It can be seen that the coherence time is $\sim 10^3$ times longer than the galactic-halo coherence time $\tau_a = Q_a / (\pi \nu_a) \sim 10^6 / \nu_a$ due to the much smaller velocity spread of the Earth-bound ALP. A more precise estimate of the coherence time can be made by computing the eigenenergies numerically from the Earth's gravitational potential, as shown in Figure 2.6. The relative frequency spacing of the lowest 7 eigenstates is $\sim 10^{-9}$ or 1 ppb, which is consistent with the rough estimate in Equation (2.43).

To determine the exact coherence time, numerical solutions of Equation (2.11) and simulations of interference between eigenstates are needed. When the ALP mass or Compton frequency is small, the coherence time can be very long, which may exceed the relaxation times of the NMR sample and the duration of a typical CASPER-Gradient measurement. In such scenarios, the ALP field will result in a one-bin signal in the frequency spectrum, and the amplitude of the signal will be limited by the relaxation times of the NMR sample. Therefore, the exact value of the coherence time is not always needed for the analysis of the CASPER-Gradient data.

2.2.6 Periodic modulations

The Earth-bound ALP wavefunction ψ_{nlm} is defined in the Earth-centered frame introduced above. A Cartesian coordinate system is chosen with the quantization axis, $\hat{z}$, along the Earth–Sun line, the $\hat{x}$ -axis along Earth’s instantaneous velocity in the Solar System, and $\hat{y} = \hat{z} \times \hat{x}$. Since the gravitational potential is spherically symmetric, the choice of coordinate axes is arbitrary. This frame is generally not aligned with the Earth’s rotation axis. As the Earth rotates, the orientation of the component of ∇a at any given location therefore exhibits a daily modulation. Earth’s orbital motion produces an annual modulation of the gradient component at a given location. Examples of daily and annual modulations are illustrated in Figure 2.7 and Figure 2.8, which show the daily and annual modulations of the gradient component for the $|n = 2, l = 1, m = 0\rangle$ state at the location of CASPER-Gradient.

Since the gravitational potential is assumed to be spherically symmetric, the eigenstates with the same n and l but different m are degenerate. Though every eigenstate experiences periodic modulations, the modulation may not be seen in the measurements due to the degeneracy, unless the degeneracy is lifted by, for example, lunar or solar gravity. Periodic modulations are not considered in the analysis of the CASPER-Gradient data in this thesis.

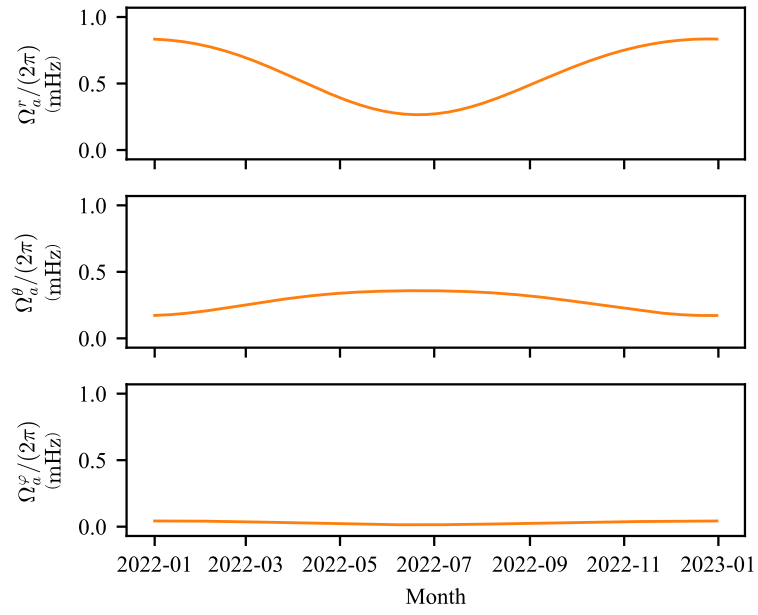


Figure 2.8: Annual modulation of the gradient component for the $|n = 2, l = 1, m = 0\rangle$ state at the location of CASPER-Gradient (Helmholtz Institute Mainz). The modulation is due to the Earth's orbit around the Sun. The Lorentz boost is included in the calculation.

Some material in this chapter is adapted from publications by the author and collaborators. The apparatus description is adapted from Ref. [59]. Sections 3.2–3.3 are partially adapted from Ref. [68] and Ref. [7].

The CASPER-Gradient-LowField apparatus converts a nuclear-spin signal into a detectable electrical output through three functional blocks: the magnet system sets the operating bias magnetic field and thereby the Larmor frequency of the sample; the sample and pickup coil generate and capture the magnetic response; and the SQUID and lock-in amplifier chain convert, amplify, and record the signal. This chapter describes each block, summarizes the key parameters, and then develops the strategy used to scan an ALP-mass range efficiently.

3.1 Apparatus

3.1.1 Cryostat and magnet

The apparatus is built around a flow-through LHe cryostat inside a Faraday room, shielded from external electric fields. It consists of an outer shell, reservoirs for liquid nitrogen (LN_2) and LHe, and vacuum insulation layers in between. LN_2 cools the cryostat and reduces the evaporation of LHe, while the superconducting devices and niobium (Nb) shield are submerged in the LHe bath at 4.2 K. When fully filled, the cryostat can maintain a superconducting environment for approximately one week. The Nb shield repels external magnetic fields (Meissner effect), isolating internal devices from external magnetic noise. Additional shielding is provided by a μ -metal shield outside the cryogenic reservoir and by a Faraday cage enclosing the entire apparatus, which suppresses electromagnetic interference (EMI).

The cryostat houses a superconducting solenoid that can generate a bias magnetic field B_0 up to 0.1 T. The DM mass range accessible to the experiment depends on the gyromagnetic ratio of the sample and on B_0 ; for proton spins, the covered Larmor frequency range goes up to 4.3 MHz.

Surrounding the solenoid, 8 superconducting shim coils (named Z1, Z2, X, Y, XY, XZ, YZ, and X^2 - Y^2) are used to reduce the field inhomogeneity to less than 2 ppm

over a spherical volume with a radius of 4 mm. The shim coil channel names indicate the geometry of the magnetic field they generate, enabling systematic correction of field inhomogeneities. In the measurements reported here, an inhomogeneity of approximately 10 ppm was achieved after tuning the shim fields using a Bayesian-optimization algorithm [58]. The larger-than-design inhomogeneity results from insufficient shimming duration, use of a longer sample (24 mm in height), and the presence of superconducting coils.

Helmholtz coils wound outside the cryostat provide a controllable offset on B_0 , enabling fine-tuning of the Larmor frequency without ramping the main solenoid, which is time-consuming and can change the magnetic field by an unknown amount. This arrangement is used to step the Larmor frequency in increments of a few hertz during the scan.

A variable-temperature insert (VTI) positions the sample in the homogeneous region of the bias magnetic field and thermally isolates it from the cryogenic bath by vacuum insulation. The SQUIDs and the fiberglass coil stacks on which Helmholtz coils, excitation coils, and the pickup coil are mounted are attached to the outer surface of the VTI.

A schematic of the cryostat and superconducting devices is shown in Figure 3.1.

3.1.2 Excitation coils

To perform pulsed-NMR schemes, an excitation coil made of Nb wires is positioned near the sample to avoid Johnson–Nyquist noise from thermal agitation in normal-conducting materials. The excitation coil provides B_x , B_y , B_z , and dB_z/dz channels, with the first three capable of producing homogeneous fields along the respective axes and the last generating a linear gradient along the z -axis.

Radio-frequency (rf) pulses are generated by a function generator and amplified by an amplifier. A tuning circuit composed of capacitors and inductors is designed for impedance matching between the function generator and the excitation coil across different frequencies.

3.1.3 Sample and sample insert

Liquid methanol (CH_3OH) was chosen as the sample because of its high proton density (99 mmol/cm^3) and low freezing point (175.6 K), which allows it to remain liquid at the operating temperature of approximately 180 K. With four protons per molecule, the 1.2 cm^3 cylindrical sample (radius 4 mm, height 24 mm) contains 7.2×10^{22} protons.

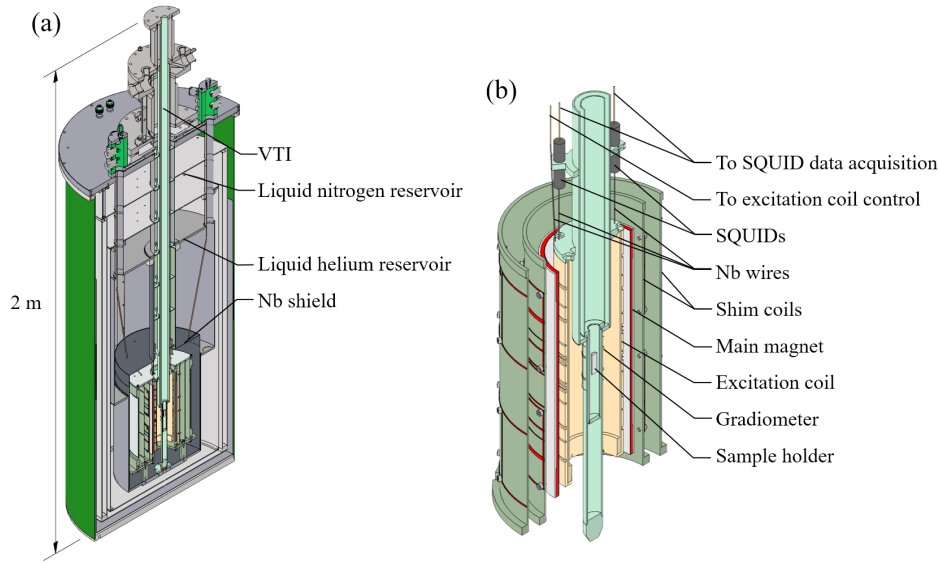


Figure 3.1: (a) Cryostat for maintaining superconducting operating temperatures. The LHe bath provides a 4.2 K environment for the superconducting apparatus. (b) Superconducting apparatus including the main coil, shim and excitation coils, SQUIDs with a Nb capsule, Nb connecting wires, and a pickup coil in first-order gradient configuration made of Nb wires.

The sample holder is made of PEEK, a polymer with low magnetic susceptibility and good thermal conductivity at cryogenic temperatures. The holder is positioned in a sample insert, which is a long PEEK tube that can be inserted into the VTI. The sample temperature is regulated by flowing warm nitrogen gas through a spiral surrounding the sample holder. Temperature stability can be monitored by repeated pulsed-NMR measurements: a constant NMR linewidth indicates a stable temperature. A schematic of the sample and sample insert is shown in Figure 3.2.

At thermal equilibrium in the bias magnetic field B_0 , the proton spins acquire a net longitudinal magnetization

$$M_0 = \frac{n \gamma^2 \hbar^2 I(I+1) B_0}{3 k_B T}, \quad (3.1)$$

where $n = 6.0 \times 10^{22} \text{ cm}^{-3}$ is the proton number density of liquid methanol, $\gamma = 2\pi \times 42.577 \text{ MHz/T}$ is the proton gyromagnetic ratio, $I = 1/2$ is the proton nuclear spin, k_B is the Boltzmann constant, and T is the sample temperature. At

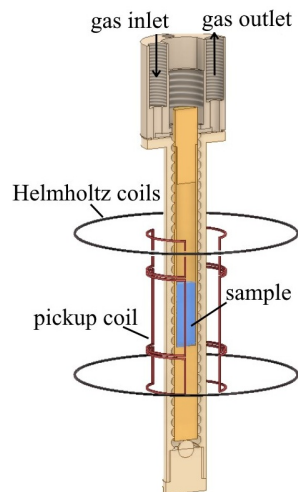


Figure 3.2: The methanol sample (blue) sits at the center of the three-coil Nb saddle-pair pickup (red). Helmholtz coils (black) fine-tune the bias magnetic field. Adapted from Ref. [59] with permission.

$B_0 = 31.67$ mT and $T \approx 180$ K, the thermal polarization is $\approx 1.8 \times 10^{-7}$ and $M_0 \approx 1.5 \times 10^{-4}$ A/m = 1.9×10^{-10} T/ μ_0 [67].

3.1.4 Pickup coil

A Nb pickup coil mounted on the outside of the VTI in the LHe bath detects the transverse magnetization of the sample. The coil consists of three saddle-pair coils in a gradiometer configuration: the area of the central coil is twice that of the two outer coils, and the coils are wound in opposing senses so that far-field flux contributions cancel. The height of the central coil is 24 mm, while the two outer coils are 12 mm high. The radius of each coil is 14 mm. The opening angle of each saddle-pair coil is 120° . The filling factor—the fraction of the coil volume occupied by the sample—is approximately 8 % [67]. The pickup coil geometry and pulsed-NMR scheme are illustrated in Figure 3.2.

3.1.5 SQUID detector

Magnetic flux in the pickup coil is coupled into a direct-current superconducting quantum interference device (dc-SQUID), as shown in Figure 3.4. The SQUID circuit is enclosed in a Nb capsule for shielding and protection from external disturbances. The SQUID converts magnetic flux into electrical signals through the com-

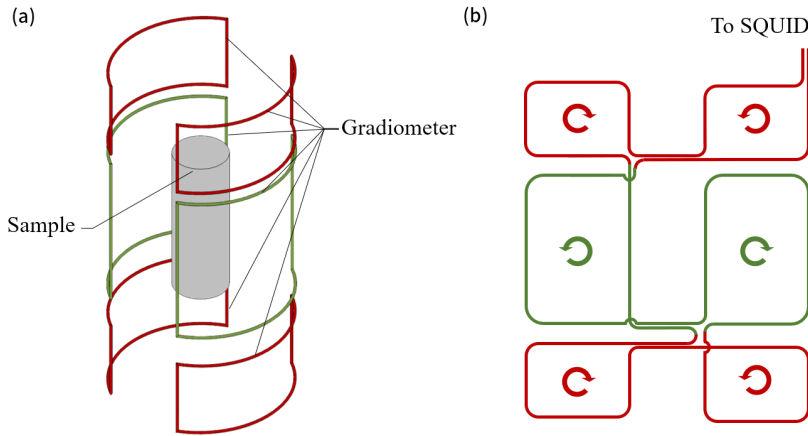


Figure 3.3: (a) Simplified schematic of the gradiometer with the sample positioned at its center. (b) Details of the gradiometer loop connections showing the three-coil saddle-pair configuration with opposing windings for noise rejection.

bination of flux quantization and Josephson tunneling effects, offering exceptional sensitivity for weak magnetic signals [43]. The SQUID is operated in flux-locked-loop (FLL) mode during measurements, i.e., a feedback voltage is applied to the SQUID to maintain the magnetic flux in the coils so that it remains at the linear operating point of its $V - \Phi$ characteristic, as illustrated in Figure 3.4. The feedback voltage is the measured output.

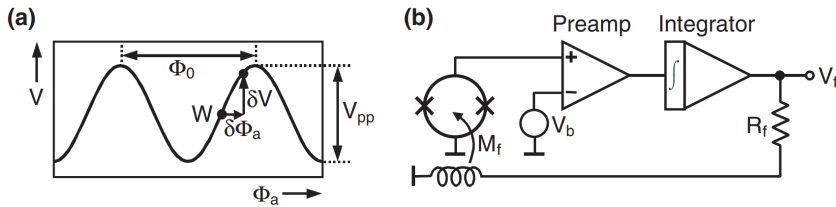


Figure 3.4: (a) $V - \Phi$ characteristic of a dc-SQUID showing the operating point W where the feedback voltage has an approximately linear dependence on magnetic flux. (b) Directly coupled FLL circuit with built-in pre-amplifier and integrator for maintaining the SQUID at the linear operating point. Reproduced from [0] with permission.

The feedback voltage V_f is related to the magnetic flux in the pickup coil Φ_{pick} by

$$V_f = \frac{R_f}{M_f} \Phi_{\text{in}} = \frac{R_f}{M_f} \frac{M_{\text{in}}}{L_{\text{pick}} + L_{\text{in}}} \Phi_{\text{pick}} = \varepsilon \Phi_{\text{pick}}, \quad (3.2)$$

where R_f is the feedback resistance, M_f is the feedback-coil mutual inductance, Φ_{in} is the input magnetic flux in the SQUID, M_{in} is the mutual inductance between the pickup and SQUID input coils, and L_{pick} , L_{in} are the respective coil inductances. The relevant parameters are summarized in Table 3.1.

Table 3.1: Properties of the dc-SQUID sensor used in the measurements [67].

Pickup coil inductance (L_{pick})	553 nH
SQUID input coil inductance (L_{in})	400 nH
Mutual inductance (M_{in})	3.98 nH
Feedback-coil inductance (M_f)	$22.83 \Phi_0/\text{mA}$
Feedback resistance (R_f)	3 k Ω
Conversion factor (ε)	$0.549 \text{ mV}/\Phi_0$

The detection chain is also characterized by the transfer coefficient β , defined as the ratio of the SQUID feedback-voltage amplitude $V_{\perp}$ to the transverse sample magnetization $\mu_0 M_{\perp}$:

$$\beta = \frac{V_{\perp}}{\mu_0 M_{\perp}}. \quad (3.3)$$

The transfer coefficient was determined at each scan step from the feedback voltage following a $\pi/2$ pulse, for which $M_{\perp} = M_0$. The average value is $\beta \approx 4 \text{ mV/nT}$.

The SQUID output is fed into a Zurich Instruments MFLI Lock-in Amplifier (LIA), which demodulates the signal into in-phase (I) and quadrature (Q) channels. The outputs of the two channels are recorded at a sampling frequency of 13.4 kHz and are used to generate spectra via fast Fourier transformation (FFT).

During ALP-search measurements, the SQUID and FLL electronics are battery-powered to avoid electromagnetic interference from the mains. All other electronics (lock-in amplifier, field-ramp supply) are outside the Faraday cage; only shielded Bayonet Neill-Concelman (BNC) cables penetrating the cage are used to carry signals.

3.1.6 Experimental control and data acquisition

Comprehensive centralized control of the CASPER-Gradient-LF apparatus is implemented in Python to coordinate the complex timing requirements of pulsed-NMR

experiments and manage communication between diverse hardware components. This control system enables synchronized operation of shim coils, rf pulse generation, and SQUID data acquisition.

The main functions of the control system include:

1. Magnet and shim control:

- a) Remote switching of the shim power supply on and off
- b) Connection and disconnection of communication between the power supply and the shim coils
- c) Real-time setting of shim currents to desired values, with optional heater control for current stabilization

2. Pulse generation and timing:

- a) Command interface for injecting radiofrequency pulses
- b) Recording rf pulse timing using an oscilloscope and storing it in HDF5 format
- c) Analysis of function generator data and pulse characterization

3. Data acquisition and processing:

- a) Recording SQUID data and automated creation of HDF5 files for storage
- b) Real-time analysis and monitoring of acquired signals
- c) Generation and analysis of power spectral densities

The control system orchestrates multiple measurement schemes, including pulsed-NMR diagnostics for sample characterization, automated shimming procedures with feedback optimization, ALP measurements, and magnetic field monitoring. Additionally, the system includes capabilities for NMR kinetic simulations, stochastic ALP field simulations, computation of apparatus properties, and synchronized file management across multiple terminals.

3.2 Sensitivity

The transverse magnetization induced by the ALP field couples to the pickup coil and SQUID. Due to the stochastic nature of the ALP field, the signal can vary significantly over time and is therefore characterized by its average power rather than

a deterministic amplitude. In addition to the ALP coherence time τ_a , the signal is also affected by the sample relaxation time T_2^* . When $T_2^* \ll \tau_a$, the amplitude of the signal, characterized by the tipping angle ξ , is limited by the sample relaxation time, and it grows as

$$\xi(t) = (1 - e^{-t/T_2^*})\Omega_a T_2^*, \quad (3.4)$$

and saturates at $t \gg T_2^*$. When $T_2^* \gg \tau_a$, the signal first grows linearly at $t \ll \tau_a$, and then grows as $\sim (t\tau_a)^{1/2}$ at $\tau_a < t < T_2^*$ like a random walk [31, 68], saturating at $t \gg T_2^*$. An analytical expression for the signal in this regime derived from numerical simulations is

$$\xi(t) = \left(1 - \exp\left(\frac{-t}{\frac{1}{2}T_2^*}\right)\right)^{1/2} \Omega_a (T_2^* \tau_a)^{1/2}. \quad (3.5)$$

To unify the two regimes, we introduce a parameter x that interpolates between the two limits:

$$\xi(t) = \left(1 - \exp\left(\frac{-t}{xT_2^*}\right)\right)^x \Omega_a T_2^* \left(\frac{\tau_a}{\tau_a + T_2^*}\right)^{1/2}. \quad (3.6)$$

For most measurements, $T_2^*, \tau_a \ll T$ holds, so the ALP-induced signal amplitude is approximately

$$\xi = \Omega_a T_2^* \left(\frac{\tau_a}{\tau_a + T_2^*}\right)^{1/2}. \quad (3.7)$$

This tipping angle corresponds to a transverse magnetization $M_\perp = M_0 \sin \xi \approx M_0 \xi$ for small ξ , which is the case for the ALP-induced signal considering the expected Rabi frequencies in Equations (2.2) and (2.37). Taking Equations (3.3) and (3.7) into account, the following relationship is obtained:

$$V_a = \beta \mu_0 M_0 \Omega_a T_2^* \left(\frac{\tau_a}{\tau_a + T_2^*}\right)^{1/2} \propto g_{\text{aNN}} T_2^* \left(\frac{\tau_a}{\tau_a + T_2^*}\right)^{1/2}, \quad (3.8)$$

where V_a is the SQUID feedback voltage corresponding to the ALP-induced transverse magnetization. In the following discussion, the experimental parameters β and M_0 are treated as constants and therefore omitted for simplicity, and the ALP-induced signal is expressed as $g_{\text{aNN}} T_{\text{coh}}$ where

$$T_{\text{coh}} = T_2^* \left(\frac{\tau_a}{\tau_a + T_2^*}\right)^{1/2}. \quad (3.9)$$

is the effective coherence time of the ALP-induced signal.

The signal amplitude in the power spectral density (PSD) is

$$\text{PSD} = \frac{(g_{\text{aNN}} T_{\text{coh}})^2}{\Delta\nu}, \quad (3.10)$$

where $\Delta\nu$ is the expected signal linewidth, which is determined by the narrower of the ALP linewidth $\Delta\nu_a = 1/(\pi\tau_a)$ and the NMR linewidth $\Delta\nu_n = 1/(\pi T_2^*)$. Hence, $\Delta\nu$ can be expressed as

$$\Delta\nu = \frac{1}{\pi} \frac{1}{\tau_a + T_2^*}. \quad (3.11)$$

Within a spectral bin of width $\Delta\nu$, the number of PSD values available for averaging from a measurement of duration T is

$$N_{\Delta\nu} = T\Delta\nu \gg 1, \quad (3.12)$$

assuming that the measurement duration is much longer than the ALP coherence time and the sample relaxation time. The noise is characterized by its standard deviation,

$$\sigma = \sqrt{\frac{1}{T\Delta\nu}} \sigma_0, \quad (3.13)$$

where σ and σ_0 are the noise standard deviations with and without averaging, respectively. Since $N_{\Delta\nu} \gg 1$, the noise distribution approaches a Gaussian distribution according to the central limit theorem. The coupling strength g_{aNN} corresponding to a 5σ signal can be found using the signal-to-noise ratio (SNR) condition

$$\frac{\text{PSD}}{\sigma} = 5. \quad (3.14)$$

which can be expanded using Equations (3.10), (3.11), and (3.13) to

$$\frac{g_{\text{aNN}}^2 T_{\text{coh}}^2 T^{1/2}}{\Delta\nu^{1/2} \sigma_0} = 5. \quad (3.15)$$

Hence, the g_{aNN} sensitivity is

$$|g_{\text{aNN}}| \propto T^{-1/4} (T_2^*)^{-1} \tau_a^{-1/2} (T_2^* + \tau_a)^{1/4}. \quad (3.16)$$

Here a constant factor of $5^{1/2} \sigma_0^{1/2} \pi^{-1/4}$ is dropped since only the scaling of g_{aNN} with τ_a , T_2^* , and T is of interest. In terms of the two extreme regimes of τ_a relative

to T_2^* and T_2 , the sensitivity can be expressed as

$$|g_{\text{aNN}}| \propto \begin{cases} T^{-1/4} (T_2^*)^{-1} \tau_a^{-1/4}, & T_2^* \ll \tau_a \\ T^{-1/4} (T_2^*)^{-3/4} \tau_a^{-1/2}, & T_2^* \gg \tau_a \end{cases}. \quad (3.17)$$

The implications of Equation (3.17) can be summarized as follows:

1. When the measurement duration T is much longer than the sample relaxation time T_2^* and the ALP coherence time τ_a , the sensitivity is dominated by the term $T^{-1/4}$, indicating a slower improvement with increasing measurement duration.
2. The sensitivity improves with a longer sample relaxation time T_2^* . The improvement is more significant when $T_2^* \ll \tau_a$.

3.3 Scanning

3.3.1 Procedure

The CASPER-Gradient-LowField experiment scans the ALP-mass parameter space by sweeping the magnitude of B_0 , which tunes the Larmor frequency of the sample to a series of values $\nu_i \in [\nu_{\text{start}}, \nu_{\text{end}}]$. One step of the scan consists of:

- A. Pulsed-NMR measurement to determine the resonance frequency and the relaxation time T_2^* .
- B. Spin-echo measurement to determine T_2 .
- C. Data acquisition with the SQUID magnetometer—no transverse pulse applied—to record the ALP-signal channel over the measurement duration T .
- D. Ramp B_0 to the next value.

The frequency scan-step size is set by the sensitive bandwidth of a single step. The sensitive bandwidth is a bandwidth around the Larmor frequency where the experiment is approximately equally sensitive to ALP signals. It can be estimated from the NMR linewidth, the ALP linewidth, and the noise profile. For non-amplifiable noise (white noise with a flat spectrum), the experiment is sensitive over a bandwidth approximately equal to the maximum of the NMR linewidth and the ALP linewidth. For noise amplified by NMR, the experiment is equally sensitive over a

bandwidth much larger than the NMR linewidth. Choosing a step size larger than the sensitive bandwidth leaves gaps in sensitivity; choosing a smaller step size multiplies the number of steps and the dead time between steps without improving sensitivity. Therefore, the scan-step size is set to be approximately equal to the sensitive bandwidth of a single step. Examples of amplifiable and non-amplifiable noise are illustrated in Figure 3.5. For CASPER-Gradient experiments, the dominant noise type is non-amplifiable noise, which is assumed for the following discussion.

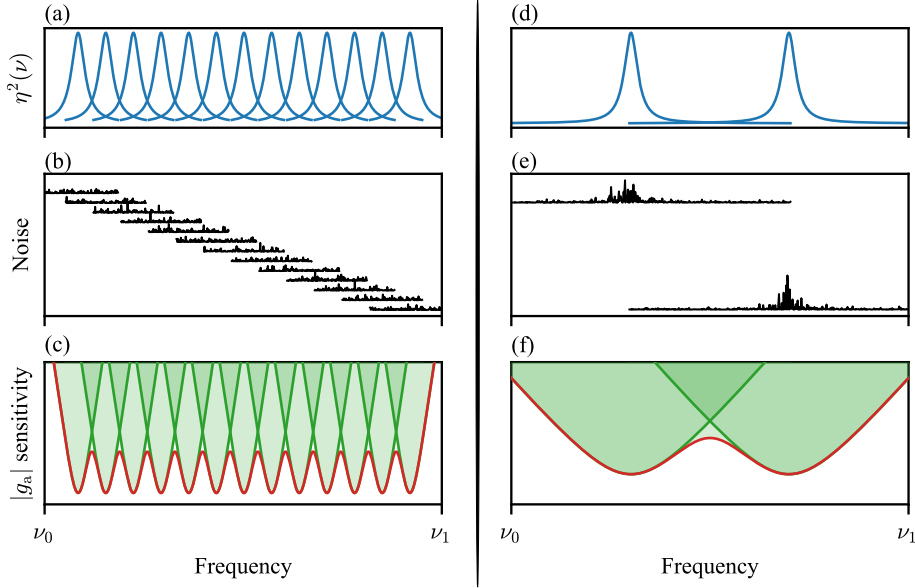


Figure 3.5: Examples of experiments scanning over the frequency range $[\nu_0, \nu_1]$. Each of the spectra corresponds to one scan step over a certain frequency range. Left column: Dominance of non-amplifiable noise. Right column: Dominance of amplifiable noise. Rows (top to bottom): Frequency spectra of the square of magnetic signal amplification $\eta(\nu)$ in panels (a) and (d), PSD noise in panels (b) and (e), and the corresponding ALP coupling strength g_a sensitivity in panels (c) and (f). Offsets have been added to the noise spectra for clarity. The scan-step size is larger when amplifiable noise dominates due to the broader sensitivity bandwidth. Adapted from Ref. [7], originally prepared by the author.

3.3.2 Optimization of scanning

Two questions arise after the previous discussion: What is the figure of merit for a scanning strategy? How can the scanning strategy be optimized to maximize

the figure of merit? The choice of figure of merit is arbitrary, but a natural choice is the area covered in the $\log |g_{\text{aNN}}| - \log \nu$ mass-coupling parameter space. Given a fixed total measurement time, the relaxation time T_2^* can be tuned to maximize the parameter space coverage. A short relaxation time T_2^* may allow for a larger scan-step size and fewer steps in total, and therefore increases the measurement time per step. However, as can be estimated from the scalings in Equation (3.17), the increase in the measurement time per step is not sufficient to compensate for the decrease in relaxation time in terms of sensitivity.

A more careful analysis of the trade-off between relaxation time and measurement time per step is given as follows. With a fixed total measurement time T_{tot} , the measurement duration can be chosen to be proportional to the ALP coherence time: $T \propto \tau_a \propto \nu^{-1}$.⁶ The scan rate can be found as

$$\frac{d\nu}{dt} = \frac{\Delta\nu}{T} = \frac{\Delta\nu}{\kappa\tau_a}, \quad (3.18)$$

where κ is a constant to be determined later from the total measurement time and the scan range. The two regimes in Equation (3.17) are discussed separately. In the $T_2^* \ll \tau_a$ regime, $\Delta\nu = 1/(\pi\tau_a)$. A shorter T_2^* does not allow for a larger scan-step size, so the sensitivity is worse when T_2^* is shorter. In the other regime where $T_2^* \gg \tau_a$, $\Delta\nu = 1/(\pi T_2^*)$. Using $\tau_a = Q_a/\nu$ and Equation (3.18), the scan rate becomes

$$\frac{d\nu}{dt} = \frac{\nu}{\pi\kappa Q_a T_2^*}. \quad (3.19)$$

which leads to

$$\int_{\nu_{\text{start}}}^{\nu_{\text{end}}} \nu^{-1} d\nu = \int_0^{T_{\text{tot}}} \frac{1}{\pi\kappa Q_a T_2^*} dt. \quad (3.20)$$

Solving for κ gives

$$\kappa = \frac{T_{\text{tot}}}{\pi Q_a T_2^*} \left[\ln \left(\frac{\nu_{\text{end}}}{\nu_{\text{start}}} \right) \right]^{-1} \propto T_{\text{tot}} (T_2^*)^{-1}. \quad (3.21)$$

Using Equations (3.17) and (3.21) and omitting constant factors, the sensitivity in this regime is

$$|g_{\text{aNN}}| \propto T_{\text{tot}}^{-1/4} (T_2^*)^{-1/2} \tau_a^{-1/2}. \quad (3.22)$$

⁶ Measurement time can also be allocated uniformly across frequency steps, which makes no significant difference in the final sensitivity. A quantitative comparison of the allocation schemes is given in Ref. [68].

From Equation (3.22), the sensitivity still improves with longer T_2^* , which is consistent with the estimate in the preceding discussion.

In summary, the optimal strategy for a frequency-scanning ALP search is to work at the maximum achievable T_2^* .

Data analysis procedures and searches for Milky Way ALPs are adapted from Ref. [59], a publication by the author and collaborators. The analysis framework was developed using the data analysis pipelines of HAYSTAC [14], SHAFT [30], and CASPER-Electric [5] as references.

This chapter first presents the data analysis pipeline, including data processing and candidate validation. The data analysis pipeline is later applied to two proof-of-principle ALP searches using the CASPER-Gradient-LF apparatus with a thermally polarized methanol sample. The results, including the analysis of ALP candidates and the exclusion limits, are presented at the end of the respective sections.

4.1 Data analysis

The data analysis of CASPER-Gradient aims at identifying potential ALP signals in the frequency spectra obtained from the measurements and setting exclusion or sensitivity on the ALP coupling in the absence of a detection.

4.1.1 Methodology

In general, the raw time-series data are first converted to the frequency domain using a Fourier transform, and then processed with filters to suppress noise. The outliers in the filtered spectra, judged by a threshold determined from Monte Carlo (MC) simulations, are later evaluated for their consistency across repeated scans. **Outliers** that survive the consistency checks are considered ALP **candidates**. If no candidate is found in a frequency band, the analysis sets an exclusion limit on the ALP-nuclear spin gradient coupling in this band.

A more detailed description of the data-analysis procedure is given below and summarized in Figure 4.1. The analysis consists of 4 main stages:

- **Frequency spectrum generation and filtering.** Recorded time-series data are checked for sanity. Data that pass these checks are Fourier transformed to generate frequency spectra. Digital filters, such as the Savitzky–Golay (SG) filter and optimal filters, are applied to remove the baseline and optimize the SNR. The filtered spectra are then searched for outliers.

- **MC simulations to determine the outlier threshold.** MC simulations with artificial ALP signals are used to determine the threshold for selecting outliers in the frequency domain.
- **NMR characterization.** The NMR sample, bias magnetic field, and detection chain are characterized by pulsed-NMR measurements. These NMR configurations are later used for examining the outliers and determining the experimental sensitivity to ALPs.
- **Outlier examination.** Based on the information from previous steps, outliers in the frequency spectra are selected and examined. The validated outliers become ALP signal candidates for further investigation. Exclusion or sensitivity plots can also be generated after the validation check.

In addition to these procedures, the analysis also includes procedures such as calibration by artificial signal injection. The technical details of these procedures are discussed as follows.

4.1.2 Procedure

Time-series sanity check

Data with Gaussian noise are expected from the SQUID and LIA in the absence of an ALP signal. In the presence of an ALP signal:

- if the ALP signal is weak compared to the noise, the data are still approximately Gaussian;
- if the ALP signal is strong, the data are no longer Gaussian, and stochastic drift with a characteristic timescale set by the ALP coherence time is expected, as illustrated in Figure 2.1.

Moreover, when recording the data, instability and fluctuations in hardware and software could compromise data sanity. Therefore, the first step in data processing is to check the sanity of the time series. The time-series are checked for

1. **Corrupted data values.** The data are first checked for NaN (not-a-number) or infinite values. Since the time-series are stored as numpy arrays, NaN and infinity are examined by `numpy.isfinite` and `numpy.isnan`.
2. **Long-term instability (“drift”).** Slow (compared with the ALP coherence time) changes of the mean or variance due to unstable SQUID flux-locked-loop operation are tested within individual scan steps.

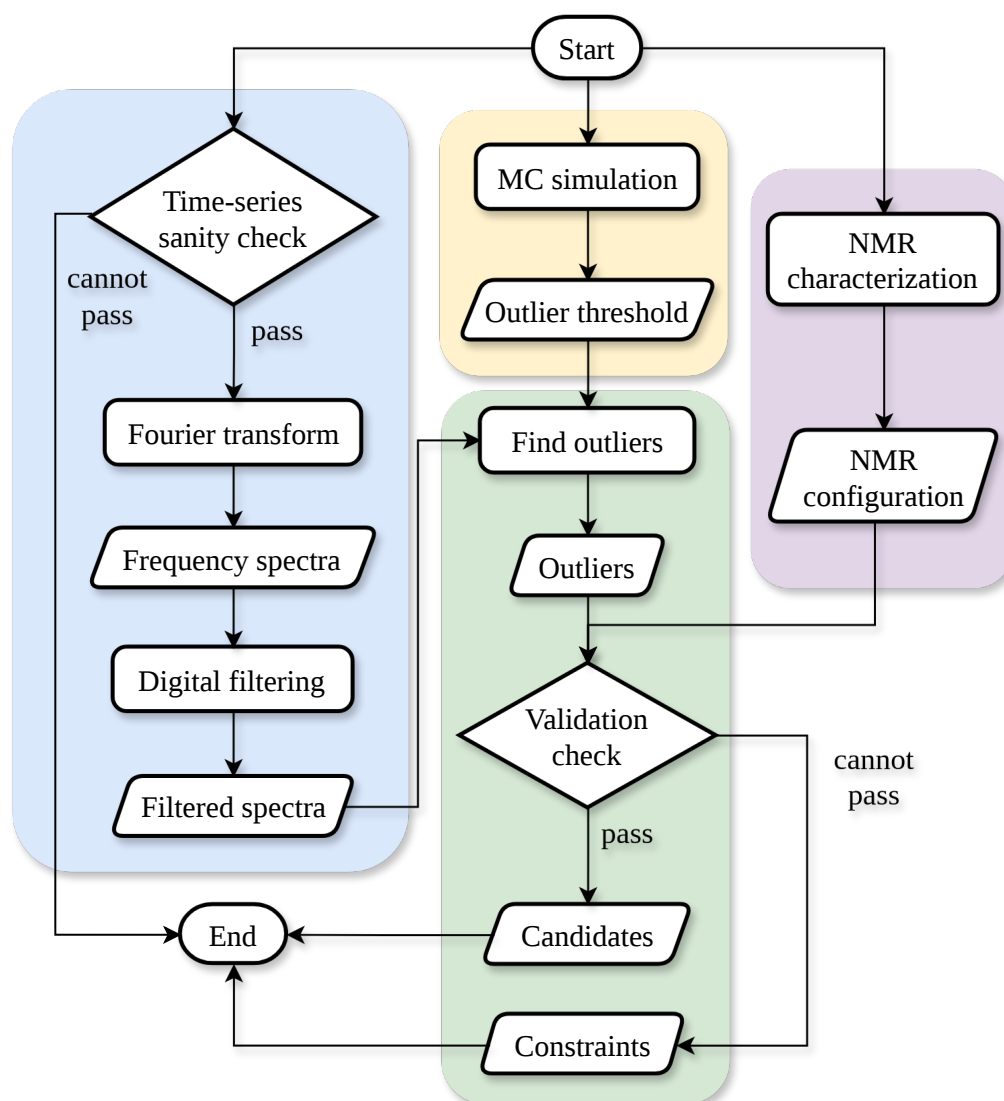


Figure 4.1: Data-analysis pipeline for the CASPER-Gradient search. Details of this pipeline are discussed in Section 4.1.2.

3. **Short-term instability (“jumps”).** Local discontinuities or clusters of outlying samples in chunks much shorter than the ALP coherence time are searched for after the drift check. Such jumps may originate from magnetic noise due to mechanical or acoustic vibration, and can produce spectral leakage and artificial outliers in the search spectrum.
4. **Non-Gaussianity.** Data without drift are tested for consistency with the Gaussian-noise assumption used in the threshold estimation.

Depending on the problems found, the following approaches may be applied:

1. **Time-series windowing.** If values in the data are found to be NaN, infinite values, jumps exceeding a certain threshold, or drifts, a window is applied to the time series to set the affected values to zero, so their effects are excluded. The window is designed to have a smooth envelope. The effects of the window on the average power of the data are later accounted for in the Fourier transform.
2. **Baseline correction.** If the drift is attributed to unstable SQUID operation, the baseline is corrected by adjusting chunks of the time series to have a common mean. Otherwise, the affected data are excluded using a window with zero values. Uncorrected drift would produce low-frequency spectral structures.
3. **Discarding.** If the fraction of invalid data exceeds a specified threshold, the corresponding measurement should be discarded. This threshold is arbitrarily chosen to be 1 %.

Note that features that do not match the ALP models investigated in this thesis, such as fluctuations on timescales much shorter than the ALP coherence time, may not be noise or faulty hardware/software behavior, but signals of other new physics. It is wise to keep track of these features for future considerations.

Fourier transform and filtering

Frequency spectra are generated by Fourier transforming time-series data that pass the sanity checks. There are several forms of the frequency spectra, including the power spectrum (PS), power spectral density (PSD), and amplitude or linear spectral density (ASD or LSD). These spectra are defined as

$$\text{PS} \propto |\text{FT}[\text{window}(t) \cdot s(t)]|^2, \quad (4.1)$$

$$\text{PSD} = \frac{\text{PS}}{\text{ENBW}}, \quad (4.2)$$

and

$$\text{ASD} = \text{LSD} = \sqrt{\text{PSD}}, \quad (4.3)$$

where FT is the Fourier transform, $\text{window}(t)$ is the window function, $s(t)$ is the time series, and ENBW is the effective noise bandwidth. In the data analysis of CASPER, ENBW is approximately the resolution bandwidth (RBW)⁷ since the applied window function is approximately rectangular. The exact formula of ENBW can be found in Equation (22) of Ref. [34]. The power spectrum is normalized in a way that the sum of the spectra is the average “power” of the signal:⁸

$$\text{Average power} = \frac{\sum_{k=0}^{N-1} |\text{window}(t_k) \cdot s(t_k)|^2}{\sum_{k=0}^{N-1} |\text{window}(t_k)|^2}, \quad (4.4)$$

where N is the number of samples in the time-series, and t_k is the time of the k -th sample. When the window function is rectangular ($\text{window}(t) = 1$ for all t), the average power is simply the sum of the power spectrum divided by the number of samples. When the window function is not rectangular, the amplification or attenuation due to the window function is corrected by the sum of the window function squared in the denominator.

After the frequency spectra are obtained, digital filters are applied for correction and SNR optimization. Specifically, these steps consist of low-pass filter attenuation correction, baseline removal, and optimal filtering.

The demodulated lock-in time-series experience digital low-pass filtering by the Zurich Instruments LIA [71]. The output time-series can be calculated using the following recursive formula

$$X_{\text{out}}[k] = e^{-T_s/T_c} X_{\text{out}}[k-1] + (1 - e^{-T_s/T_c}) X_{\text{in}}[k], \quad (4.5)$$

where X_{in} and X_{out} are the input and output time-series arrays, k is the sample index, T_s is the sampling time, and T_c is the time constant configured by the user. For the first stored sample, $X_{\text{out}}[-1]$ is chosen to be $X_{\text{in}}[0]$ by convention. This exponential moving average is also described as a digital discrete-time resistor-capacitor (DTRC) filter as it resembles the low-pass response of analog

⁷ also known as spectral resolution or bin width

⁸ Usually, power is defined as the amount of energy transferred or converted per unit time. For a voltage signal in units of V, the power is in the units of V^2/Ω . However, the “power” in the spectral analysis is defined as the square of the signal; therefore the “power” of the signal has units of V^2 . This special but common definition is adopted in this thesis unless otherwise noted.

resistor-capacitor (RC) filters. The time constant T_C determines the 3 dB bandwidth in the frequency domain. Equation (4.5) describes a first-order DTRC filter. Higher-order filters are implemented by cascading several filters in practice to suppress signals of frequencies much larger than the Larmor frequency ($|\text{frequency} - \text{Larmor frequency}| \gg \text{NMR linewidth}$). The response of the n -th-order filter in the frequency domain is approximated by the formula

$$H_n(\nu) = \frac{1}{(1 + i2\pi\nu T_C)^n}, \quad (4.6)$$

where $H_n(\nu)$ is the n -th-order attenuation at frequency ν . This attenuation is corrected by dividing the PSD by the square of the absolute value of the attenuation. If the time-series is Gaussian noise, the frequency spectra should be white noise after the low-pass filter correction, and the distribution of the PSD should follow a χ^2 distribution. Since there are two channels in the lock-in amplifier, the degrees of freedom of the distribution should be two, i.e., a χ_2^2 distribution.

Since the frequency-dependent attenuation in Equation (4.6) is only an approximation, the baseline of the PSD is not absolutely flat after the DTRC filter correction. The curve of the baseline is estimated by an SG filter and removed from the PSD. In practice, the spectrum is divided into chunks of a bandwidth much larger than the expected ALP linewidth, and a first-order SG filter is applied to each chunk to estimate the baseline.⁹ The estimated baseline is then subtracted from the spectrum, and the result is divided by the baseline to obtain the normalized power excess (NPE):

$$\text{NPE}(\nu) = \frac{\text{PSD}(\nu)}{\text{Baseline}(\nu)} - 1.$$

The next step is to decide whether optimal filtering¹⁰ should be applied or not. The optimal filter is a weighted average of neighboring frequency bins, with weights determined by the expected signal lineshape; this choice maximizes the SNR among linear filters for random noise. If the ALP linewidth is broader than the RBW of the spectrum, the signal is resolved, which means the signal is spread over multiple frequency bins. In this case, optimal filtering is applied to combine the signal power into a single bin. Alternatively, if the ALP linewidth is narrower than the RBW, the signal is unresolved and optimal filtering is not applied.

One of the procedures that was adopted in the previous CASPER-Gradient analy-

⁹ The first order is chosen so as not to affect the ALP signal. Higher-order SG filters can subtract part of the ALP signal from the spectrum.

¹⁰ also known as matched filtering

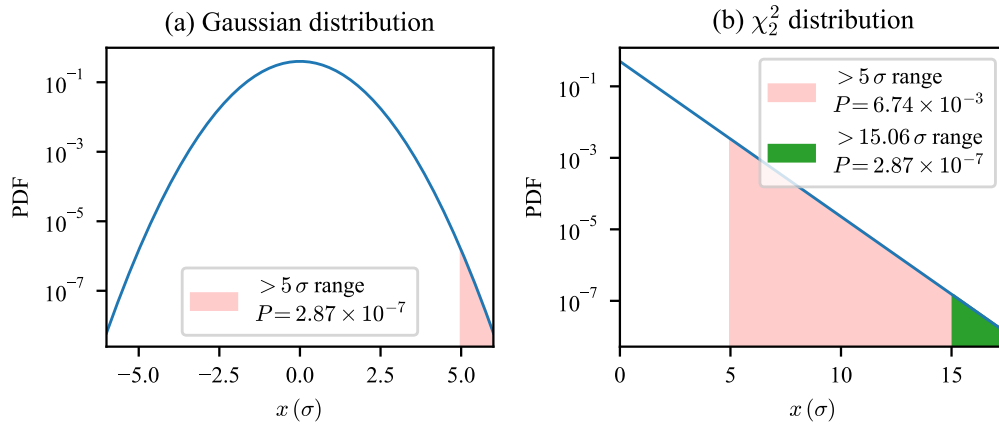


Figure 4.2: The probability density functions (PDFs) of (a) Gaussian and (b) χ^2_2 distributions. The probabilities of the shaded areas are noted in the legends.

sis [59] was the removal of narrow spikes in the frequency spectra. The philosophy behind this procedure is that spikes with linewidths much narrower than the expected ALP linewidth could be radio-frequency interference (RFI), and are unlikely to be ALP signals, so they can be removed to avoid false positives and reduce noise. However, this procedure is not applied in the analysis of the data presented in this thesis for three reasons. First, the stochastic nature of the ALP field can produce spiky ALP signals, as shown in Figure 2.1. Moreover, RFI lines are expected to be stable in frequency and amplitude, and can be identified by repeated scans. Lastly, the spikes are suppressed by optimal filtering and did not affect the outlier search in the real analysis.

After these filtering steps, the spectra are ready to be searched for outliers.

Outlier threshold

If the probability of a signal being a false positive is less than 2.87×10^{-7} , which is the probability of a Gaussian random variable exceeding five times its standard deviation σ , the signal is considered a 5σ signal. An illustration of the Gaussian distribution and the probability of a 5σ signal is shown in Figure 4.2 (a). Such signals are considered traces of new physics.

In addition to understanding what a 5σ signal is, it is also important to understand what a 5σ signal is not, and what is not a 5σ signal:

- Due to random noise, the measured amplitude of a 5σ signal fluctuates rather than remaining 5σ constantly [5, 14]. If the noise is Gaussian, and if the

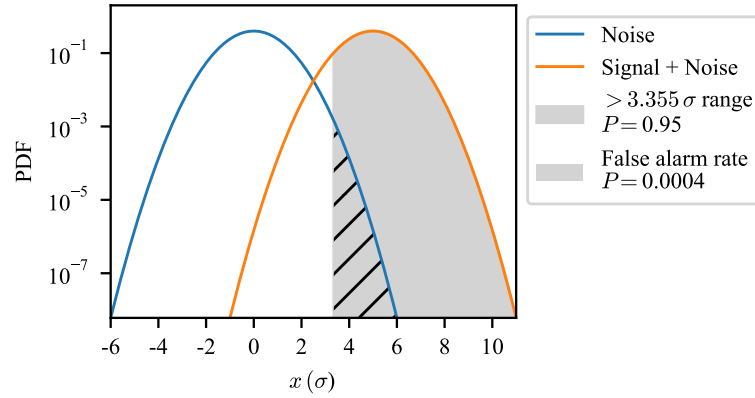


Figure 4.3: Noise and signal amplitude distributions assuming Gaussian noise and linear signal dependence.

noise is added to the signal linearly, the measured signal amplitude would be a Gaussian random variable with a mean of 5σ and a standard deviation of σ , as illustrated in Figure 4.3.

- A 5σ signal is not always a signal that exceeds five times the standard deviation of the noise, unless the noise is Gaussian. For instance, if the noise follows a χ^2_2 distribution, the threshold for a so-called 5σ signal would be 15.06σ , as shown in Figure 4.2 (b).
- 5σ signals are not rare, in terms of the number of events they represent. On one hand, 5σ signals due to non-Gaussian noise are common in real experiments. On the other hand, experiments can accumulate huge amounts of data ($\gg 1/(2.87 \times 10^{-7})$ data points), and the probability of one 5σ signal appearing due to random fluctuation is not negligible. This is also known as the look-elsewhere effect.

The exotic 5σ signals measured by CASPER would appear as peaks in the NPE spectra. To capture the 5σ signals, a threshold is set, above which frequency bins are selected as outliers for further evaluation.

The value of the threshold should not be too low or too high. If the threshold is too low, noise fluctuations are likely to be mistaken for signals (false positives). If the threshold is too high, signals are likely to be missed (false negatives). An appropriate threshold can either be determined by noise statistics or by MC simulations. A common approach is to find a threshold that captures most signals while allowing a moderate number of noise fluctuations to pass. In other words,

this threshold corresponds to a certain confidence level (C.L.), such as 95 %. For example, if noise and signal are both Gaussian random variables with the same standard deviation σ and different means (mean of noise being zero), a threshold of 3.355σ is found to ensure that, in the presence of a 5σ signal, the chance of capturing the signal is 95 %, and the chance of a false positive (false alarm rate) is only 4×10^{-4} , as illustrated in Figure 4.3. When the noise distribution is not Gaussian, the threshold can be determined by MC simulations, in which the noise and signal are simulated with the same statistics as in the experiment.

NMR characterization

The sample and the bias magnetic field are characterized by the following parameters and measurements:

1. Larmor frequency, T_2^* , NMR linewidth and lineshape, polarization, and signal power measured by the free-decay signal after a 90° pulse.
2. T_2 measured using a CPMG pulse sequence.

These parameters are used to determine the sensitivity to ALP signals in the frequency band of interest. Since these parameters may vary due to changes in the sample temperature, bias magnetic field, and other environmental factors, these measurements are performed at each scan step.

Validation check and results

Outliers that pass the threshold are not necessarily ALP signals. Several methods are used to validate the outliers. Those that can pass the validation tests are considered ALP signal candidates.

The key principle of the validation check is that ALP signals are expected to be consistent in terms of Compton frequency and coupling strength across measurements. Following this principle, the outliers are examined by the following steps:

1. **Merging outliers within individual scan steps by frequency.** The frequency of the outlier should be consistent within a certain tolerance. This tolerance is determined by the spectral RBW and the ALP linewidth. Outliers that appear at adjacent frequencies are likely to be the same ALP signal. Therefore, the first step in the validation check is to merge the outliers by their frequencies. The merged outlier is represented by the outlier with the largest amplitude.

2. **Obtaining coupling strength.** The coupling strength of the ALP signal and its uncertainty can be obtained from the amplitude of the outlier, the NMR configuration, ALP periodic modulation, and the noise in the spectra.
3. **Reproducibility.** The outlier should be reproducible across scan steps that cover the same frequency range. Though the coupling strength should be consistent, the amplitude of the outlier may vary due to the periodic modulation of the ALP signal and changes in the NMR configuration, especially the Larmor frequency. The outlier should be observed when and only when the estimated signal amplitude is significant compared to the noise level. This rule can be used to exclude outliers that are not reproducible across scans.

In the absence of ALP candidates, the analysis sets constraints on the coupling strength as the minimum ALP coupling strength that can be observed at a given confidence level. The constraint is determined by the NMR configuration and the ALP characteristics. From the NMR characterization, the SQUID magnetic flux when the sample magnetization is tipped into the transverse plane, $\Phi_{\perp}(\nu)$, can be obtained from the PSD of the free-decay signal. For a given ALP coupling strength g_{aNN} , the tipping angle ξ of the sample magnetization is determined by the Rabi frequency of the ALP field $\Omega_a \propto g_{\text{aNN}}$ (given in Equations (2.2) for Milky Way ALP and (2.37) for Earth-halo ALP) and the effective coherence time T_{coh} (given in Equation (3.9)). The PSD signal is therefore $(\xi \Phi_{\perp}(\nu))^{1/2}$. Under the assumption that ALP coupling is weak, the tipping angle is small and $\sin \xi \approx \xi$. In the absence of an $n\sigma$ ALP signal at a certain confidence level,

$$(\Omega_a T_{\text{coh}} \Phi_{\perp}(\nu))^2 \leq n\sigma, \quad (4.7)$$

solving which gives the constraint on the ALP coupling strength g_{aNN} as a function of frequency. If a candidate is observed, then the constraint at the candidate frequency is set by the coupling strength of the ALP candidate, below which a candidate was observed.

4.2 Milky Way ALP measurement

One proof-of-principle measurement was performed on December 23, 2022. This measurement is ideal for a Milky Way ALP search because it scanned the same frequency range three times and the duration of each scan step was long enough to cover multiple ALP coherence times.

4.2.1 Experiment

The search covered a frequency band of

$$1.348\,570\text{ MHz} \pm 120\text{ Hz},$$

corresponding to proton Larmor frequencies driven by a bias magnetic field $|B_0| \approx 31.67\text{ mT}$. This maps to ALP masses in a range of

$$5.577\,237\text{ neV}/c^2 \pm 496\text{ feV}/c^2.$$

Three sequential scans over the same range were performed. In each scan, there were over 40 scan steps, and the frequency step size was 6 Hz, approximately half the NMR linewidth, so that the NMR resonant windows of adjacent steps overlap.

At each frequency step, the following actions were performed in order:

1. Free-decay measurement (one $\pi/2$ excitation pulse followed by a free-decay signal) to determine the Larmor frequency, linewidth, and NMR signal power. A Lorentzian fit to the resulting PSD peak extracts these parameters.
2. T_2 measurement using a CPMG pulse sequence.
3. 100 s measurement without any pulses.
4. Frequency step: the Larmor frequency is increased by $\approx 6\text{ Hz}$ by ramping the Helmholtz coils.

Delays to allow ring-down noise to decay or the magnetization to return close to its equilibrium are applied between steps. The measurement procedure is illustrated in Figure 4.4. The experimental parameters and the relevant NMR and ALP characteristics are summarized in Table 4.1.

For the Milky Way ALP field, the expected linewidth near 1.3 MHz is approximately 1 Hz, corresponding to a coherence time of approximately 0.3 s. Thus, each 100 s acquisition spans roughly 300 coherence times and measures the time-averaged stochastic signal power.

Over the 6-hour measurement, the ALP signal power varied between scan steps mainly because the Earth's rotation changes the component of the ALP-field gradient perpendicular to the bias magnetic field. This predictable modulation, together with the measured NMR response at each step, is included when converting a power excess into an estimated coupling strength. A plot of the effect of the

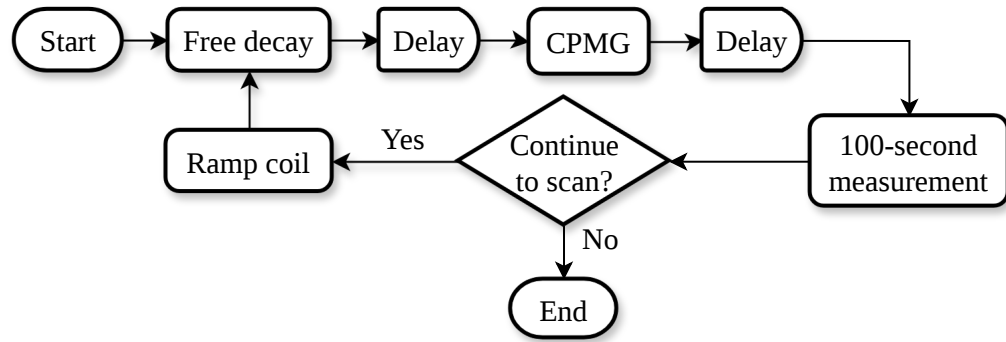


Figure 4.4: Schematic of the Milky Way ALP measurement procedure.

Earth's motion on the expected ALP signal is shown in Figure 4.5. The annual modulation due to the Earth's motion in the solar system is included in the calculation, but it did not have a significant impact over the course of the measurement.

4.2.2 Data analysis

The technical details of the data analysis, following the procedure described in Section 4.1.2, are presented.

After the time-series sanity checks, each 100 s record with windowing was Fourier transformed and corrected for the DTRC filter. The spectral baseline was estimated with a first-order SG filter and removed to form the NPE spectrum. For this search, $\text{RBW} = 1/100 \text{ s} = 0.01 \text{ Hz}$, whereas the expected ALP linewidth is approximately 1 Hz; the signal is therefore resolved over many frequency bins. The optimal filter was applied to the NPE spectrum, using the expected ALP lineshape as the filter kernel. One example of the spectra after different filtering steps is shown in Figure 4.6.

MC simulations were performed to determine the threshold for recovering a 5σ ALP signal with 95 % C.L. Artificial 5σ ALP signals were injected into simulated noise spectra before the optimal filtering, and the amplitudes of the filtered NPE were recorded. Probability distributions of the filtered NPE were obtained from many realizations of the simulated noise and signal, and the outlier threshold Θ_{RE} was determined from the distribution of the filtered NPE in the presence of a 5σ ALP signal. The simulations yielded $\Theta_{\text{RE}} = 3.43$ with a false alarm rate of approximately 0.001, as shown in Figure 4.7.

Table 4.1: Technical parameters of the Milky Way ALP measurement on December 23, 2022 (Central European Time, UTC+1).

Parameter	Name or value
Sample	thermally polarized methanol
Number of scans	3
Number of steps per scan	45
Sweep coil current increment	0.8 mA
Frequency increment per step	6 Hz
Frequency scan range	$1.348\,570\text{ MHz} \pm 120\text{ Hz}$
NMR linewidth in one step	10 Hz
T_2	0.72(5) s
ALP mass scan range	$5.577\,237\text{ neV}/c^2 \pm 496\text{ feV}/c^2$
ALP linewidth $\Delta\nu_a$	1 Hz
ALP coherence time τ_a	0.3 s
Duration per step	100 s ($\approx 300\tau_a$)
Total duration	6 h

4.2.3 Results

In order to include information about RFI and other non-white-noise structures, the data analysis searched for ALP signals in the frequency range of $1.348\,570\text{ MHz} \pm 350\text{ Hz}$, much larger than the combined NMR linewidth of $\pm 120\text{ Hz}$. This frequency range corresponds to 700 Milky Way ALP linewidths at this frequency. In the absence of an ALP signal, the number of outliers is expected to be 0.7 out of 700 based on the false alarm rate (0.1 %) obtained in the MC simulations. In the analysis, 58 outliers were found in the searched frequency range, which is much larger than the expected number of outliers due to non-white-noise structures that are not removed by the SG filters. After the validation, 6 candidates were identified among the outliers, listed in Appendix A. No candidate appeared in the NMR resonant window. Therefore, the search finds no statistically significant evidence for Milky Way ALP dark matter at the achieved sensitivity.

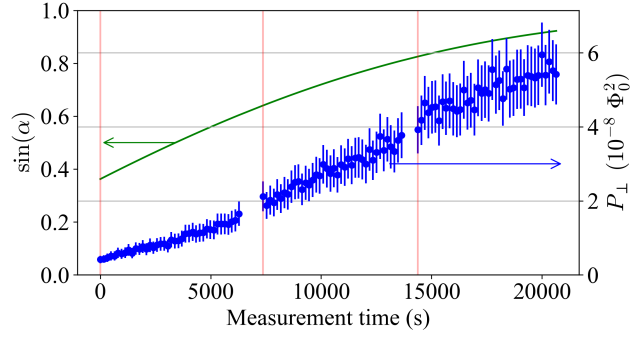


Figure 4.5: ALP parameters modulated by Earth’s motion. Plotted for the start times of each measurement step for each of the three scans in seconds. Zero time corresponds to the start of data taking on December 23, 2022, 15:47:30 (Central European Time, UTC+1). Vertical dashed red lines mark the starts of each scan. Green line: $\sin \alpha$ oscillates with a period of one day. Blue dots with error bars: expected ALP signal powers in the PSD, according to Equations (3.2) and (3.3), for the reference case of an ALP gradient coupling to protons of $g_{\text{ap}} = 10^{-9} \text{ GeV}^{-1}$. Uncertainties for these values were propagated from uncertainties in the spin relaxation times T_2 and T_2^* , sample Larmor frequency and NMR signal power. Reproduced from Ref. [59] with permission.

In the absence of a detection, an upper limit on the ALP-proton gradient coupling is obtained from Equation (4.7). In the NMR resonant bandwidth, the sensitivity is approximately $|g_{\text{ap}}| \lesssim 10^{-2} \text{ GeV}^{-1}$. The resulting exclusion limits are plotted in Figure 4.8.

The measurement sets the limit $|g_{\text{ap}}| \lesssim 10^{-2} \text{ GeV}^{-1}$ over the scanned ALP mass range $5.577\,237 \text{ neV}/c^2 \pm 496 \text{ feV}/c^2$. The sensitivity is limited by the polarization of the methanol sample; hyperpolarized samples are expected to improve it by multiple orders of magnitude.

4.3 Earth-bound ALP measurement

4.3.1 Experiment

One proof-of-principle measurement for the Earth-bound ALP halo model was performed on December 14, 2022 (Central European Time, UTC+1). This measurement is suited to the Earth-bound search because each ALP-search acquisition per step lasted 1 h, giving a narrow RBW for long-coherence-time signals.

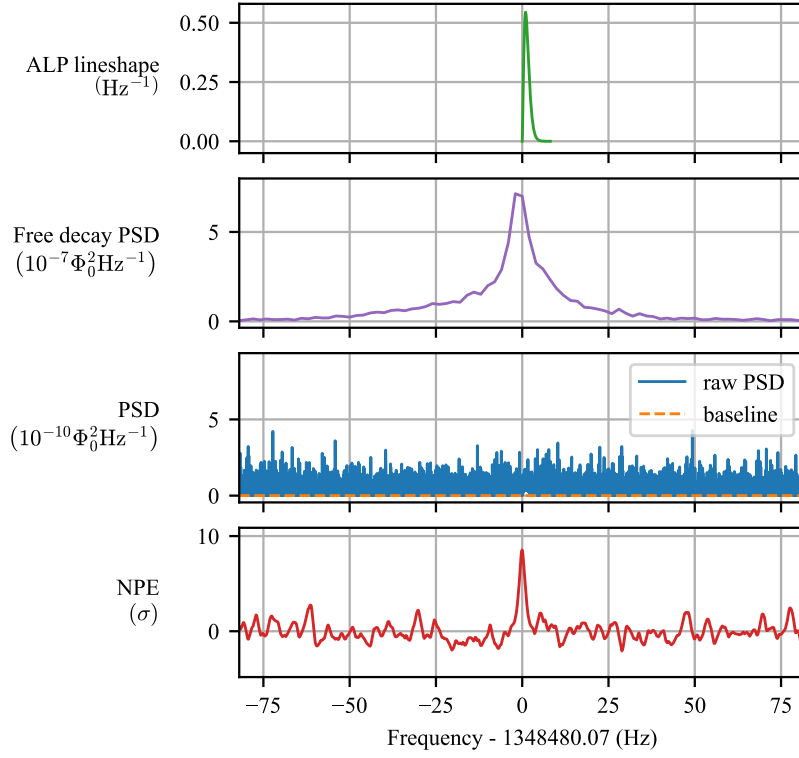


Figure 4.6: Example of data processing with an artificial ALP signal injected at the center of the spectrum. Top to bottom: lineshape of normalized ALP field, free-decay PSD, PSD of the measurement, and NPE.

The measurement scanned

$$1.348\,575\,\text{MHz} \pm 24\,\text{Hz}$$

by sweeping the magnitude of the bias magnetic field $|B_0| \approx 31.674\,03\,\text{mT}$. This maps to ALP masses in a range of

$$5.577\,258\,\text{neV}/c^2 \pm 99\,\text{feV}/c^2.$$

One scan over this range was performed. There were 8 scan steps, and the frequency step size was 6 Hz, approximately half the NMR linewidth, so that the NMR resonant windows of adjacent steps overlap.

At each frequency step, the following actions were performed in order:

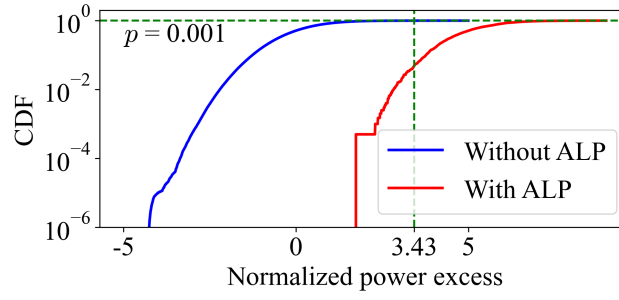


Figure 4.7: Empirical cumulative distribution function (CDF) obtained from the NPE distribution of spectra. The vertical dashed line represents the threshold for rescan, Θ_{RE} , corresponding to the null hypothesis of a 5σ ALP detection at the 95 % C.L. The p-value corresponding to the threshold (horizontal dashed line) is approximately 0.001 (false alarm rate). Reproduced from Ref. [59] with permission.

1. Set the sweep coil current.
2. Free-decay measurement (one $\pi/2$ pulse followed by a free-decay signal) to determine the Larmor frequency, linewidth $\Delta\nu$ (and hence $T_2^* = 1/(\pi\Delta\nu)$), and NMR signal power. A Lorentzian fit to the resulting PSD peak extracts these parameters and calibrates the sensitivity at each step.
3. T_2 measurement using a CPMG pulse sequence.
4. A 1-hour ALP measurement without any pulses.
5. Free-decay measurement as in step 2.
6. T_2 measurement as in step 3.

Delays ($\gg T_1$) were inserted between pulsed-NMR measurements to allow the magnetization to return close to its equilibrium value along B_0 . The measurement procedure is illustrated in Figure 4.9. Note that the 1-hour ALP measurement duration is much longer than the 100-second duration used in the data taking described in Section 4.2. The long measurement duration results in a much narrower RBW, which is beneficial for the sensitivity to ALP signals with long coherence time, such as the Earth-bound ALP halo. The experimental parameters and the relevant NMR and ALP characteristics are summarized in Table 4.2.

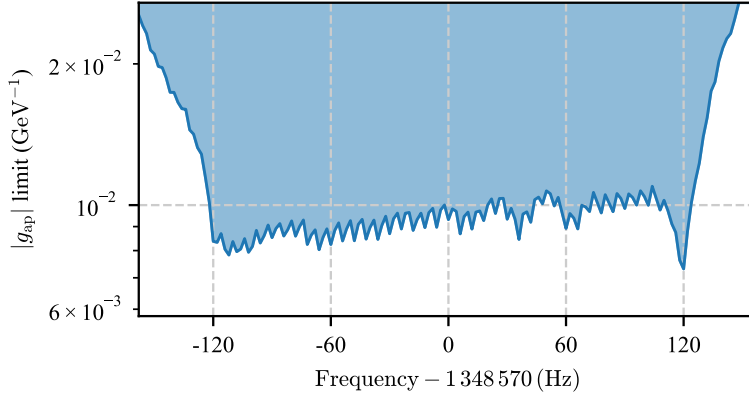


Figure 4.8: ALP gradient-proton coupling excluded (indicated by the shaded region) at 95 % C.L. by the measurement on December 23, 2022. The sensitivity is good at 1.348 690 MHz because the measurement duration was longer at this frequency. The sensitivity is worse at the edges of the scanned frequency range because the NMR response is weaker there.

4.3.2 ALP signal

As mentioned in Section 2.2.5, empirical results are presented in this section under the assumption that the Earth-bound ALPs are mostly in the lowest-energy eigenstates, where the ALP density is significant at the Earth’s surface. For the 7 lowest-energy eigenstates of the Earth-bound ALP halo, the expected daily modulation¹¹ of the ALP Rabi frequencies is shown in Figure 4.10. Since CASPER-Gradient is sensitive to the ALP-field gradient component perpendicular to the bias magnetic field, only the perpendicular component (polar and azimuthal) of the ALP Rabi frequency is relevant to the measurement. A typical value of the Rabi frequency is approximately

$$\Omega_{a,\perp}^{\oplus} \approx 2\pi \times 0.1 \text{ mHz} \frac{g_{aNN}}{10^{-9} \text{ GeV}^{-1}} \left(\frac{M_{\text{DM}}^{\text{EH}}}{4 \times 10^{-9} M_{\oplus}} \right)^{1/2} \quad (4.8)$$

during the measurement. This value is applied in the calculation of the ALP-proton gradient coupling in this section. The tipping angle of the magnetization can be estimated as $\theta \approx \Omega_a T_2 \approx 0.0002 \pi$.

The differences between the eigenfrequencies of the Earth-bound ALP halo are

¹¹ As discussed in Section 2.2.6, if no mechanism lifts the degeneracy, daily modulation would not be observed in the measurements.

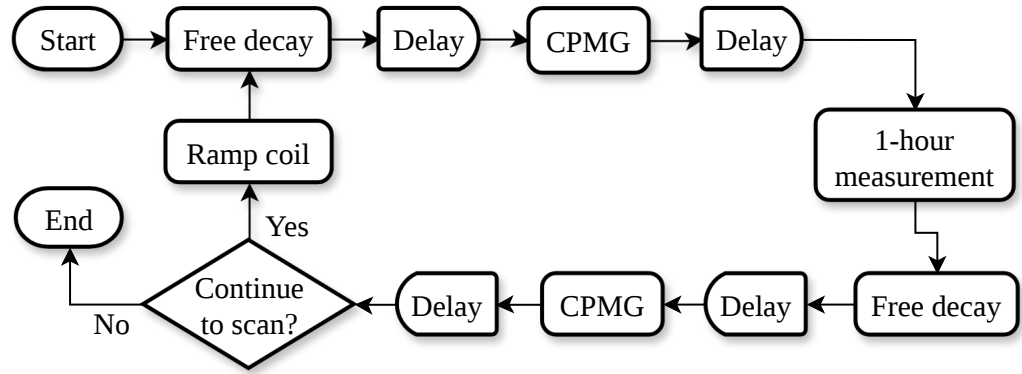


Figure 4.9: Schematic of the Earth-bound ALP measurement procedure.

on the order of 1 mHz, while the difference between the 3s and 5s states is 0.25 mHz, as shown in Figure 2.6. For the 1 h records used here, $\text{RBW} = 1/1 \text{ h} = 0.28 \text{ mHz}$, which is roughly the same order of magnitude as the ALP eigenfrequency differences. Therefore, the signal is not well resolved in the frequency spectra, and the optimal filter is not applicable for this scenario either.

4.3.3 Data analysis

The technical details of the data analysis, following the procedure described in Section 4.1.2, are presented as follows.

Short-term instabilities (“jumps”) in the time-series data were observed. The jumps are unlikely to be statistical fluctuations, since the jumps appear in clusters rather than being randomly distributed in time. One possible explanation is that the jumps are magnetic noise caused by transient mechanical vibrations. Another possible explanation is that the jumps are exotic signals, such as ALP domain walls. This thesis does not further investigate these possible transient signals of new physics.

After the jumps were removed by applying windowing, Fourier transforms were performed to generate the frequency spectra. The NPE spectrum was later formed after lock-in low-pass correction and the baseline subtraction.

The time-series signals from the two LIA channels are independent and Gaussian, so the NPE follows a χ^2_2 distribution. As discussed in Section 4.1.2, a one-sided 5σ outlier in Gaussian statistics corresponds to a p -value of 2.87×10^{-7} . This p -value corresponds to 15.06σ in the χ^2_2 distribution. In the presence of a 15.06σ ALP signal, the NPE distribution is a non-central χ^2_2 distribution. If the outlier

Table 4.2: Technical parameters of the Earth-bound ALP measurement (December 14, 2022).

Parameter	Name or value
Sample	thermally polarized methanol
Number of scans	1
Number of scan steps	8
Sweep coil current increment	0.8 mA
Frequency increment per step	6 Hz
Frequency scan range	$1.348\,575\text{ MHz} \pm 24\text{ Hz}$
NMR linewidth in one step	$\approx 10\text{ Hz}$
ALP mass scan range	$5.577\,258\text{ neV}/c^2 \pm 99\text{ feV}/c^2$
ALP linewidth $\Delta\nu_a$	$\lesssim 1\text{ mHz}$
ALP coherence time τ_a	$\gtrsim 1000\text{ s}$
T_2	$\approx 2\text{ s}$
Measurement duration per step	$1\text{ h} (\lesssim \tau_a)$

threshold is set at 7.809σ , the probability of capturing a 15.06σ ALP signal is 95 %, while the probability of a false positive (false alarm) is 4×10^{-4} . Therefore, the outlier threshold is set at $\Theta = 7.809\sigma$ for the Earth-bound ALP search. The threshold and corresponding probabilities are illustrated in Figure 4.11.

4.3.4 Empirical results

Since the expected Earth-bound ALP linewidth is narrower than the RBW, the ALP signal would not appear in more than two adjacent frequency bins. The outliers were merged if they were adjacent. After merging, 1041 outliers were found out of $\sim 720\,000$ NPE bins searched. Based on the false alarm rate, the number of outliers is expected to be around 288. The excess of outliers is likely due to non-white-noise structures such as RFI in the spectra. There was no optimal filtering to suppress these structures. Outliers appearing at consistent frequencies with consistent coupling strengths were considered candidates. Out of the 1041 outliers, 38 candidates

were found after the validation check. Note that only one scan was performed over the scan range, so the candidates cannot be cross-checked by multiple scans. Most of the candidates did not appear in the NMR resonant windows; therefore, they are unlikely to be ALP signals. Those candidates that appeared in the NMR resonant windows are likely to be aliases of RFI, since similar structures are observed in the spectra at regular intervals. The candidates are listed in Appendix A for reference.

In the absence of a detection, constraints on the ALP-proton coupling can be derived from the measured NPE noise spectra and the expected signal power for the Earth-bound ALP halo model based on Equation (4.7). At frequencies where candidates were found, the lower bounds of the constraints are set to the corresponding coupling strengths of the candidates. In the NMR resonant bandwidth, the sensitivity is approximately $|g_{ap}| \lesssim 10^{-9} \text{ GeV}^{-1}$. The resulting exclusion limits are plotted in Figure 4.12.

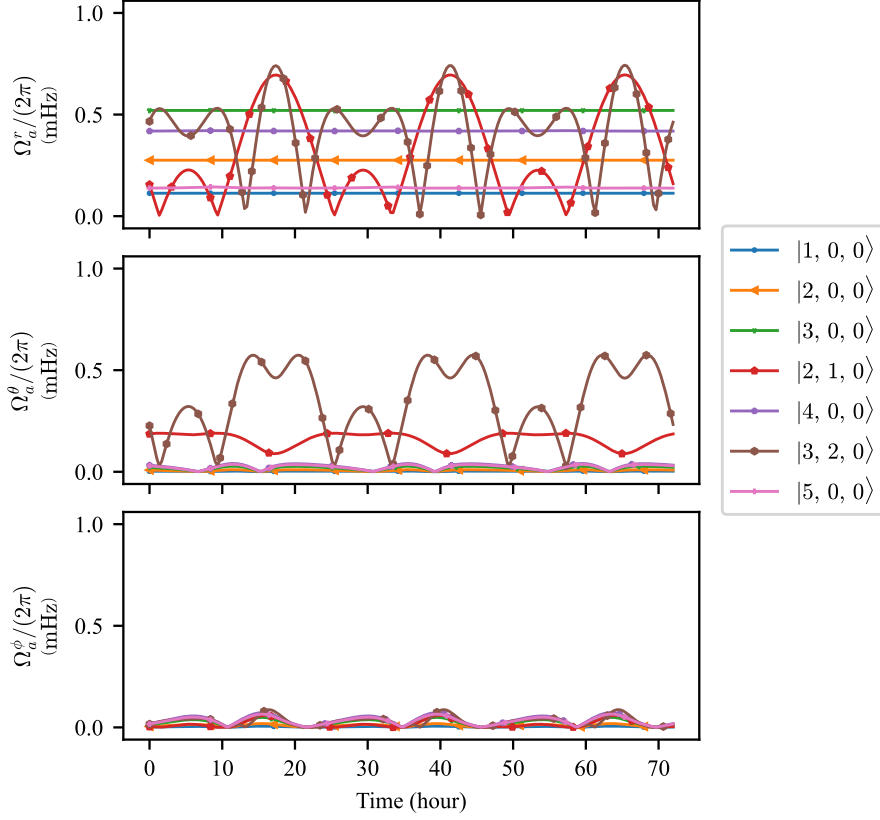


Figure 4.10: Expected daily modulation of the ALP signal for the seven lowest-energy eigenstates ($m = 0$) of the Earth-bound ALP halo, where the ALP density is significant at the Earth's surface. The ALP Compton frequency is taken as 1.348 500 MHz. The Lorentz boost is included. The quantization axis is arbitrarily chosen to be along the Earth–Sun line.

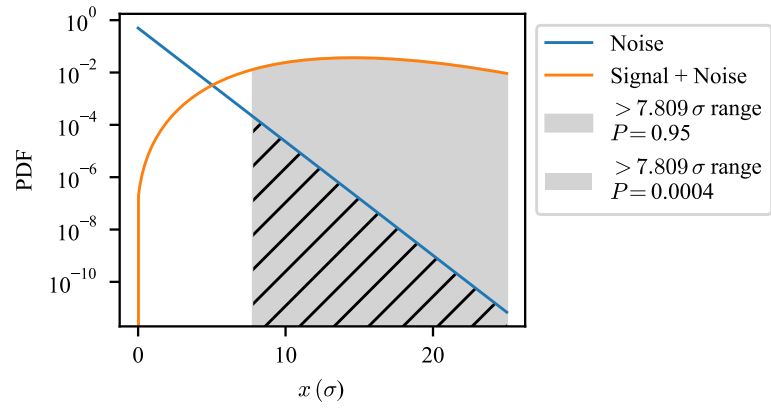


Figure 4.11: Distribution of the NPE under the null hypothesis (white noise) and alternative hypothesis (ALP signal).

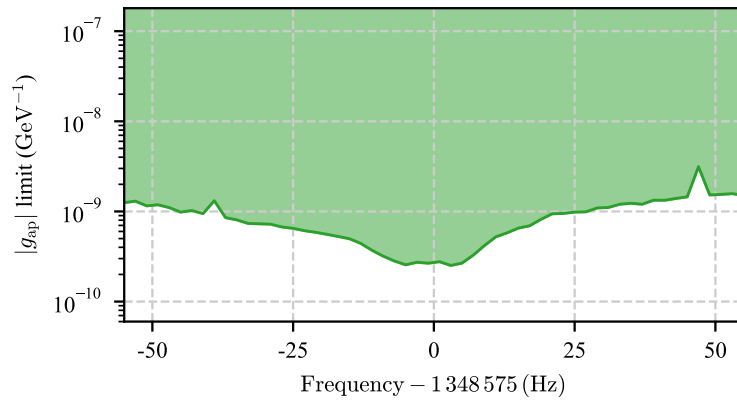


Figure 4.12: ALP gradient-proton coupling excluded at 95 % C.L. by the measurement on December 14, 2022.

5.1 Conclusions

This thesis investigated axionlike-particle dark matter through the nuclear-gradient coupling, using nuclear magnetic resonance as the resonant detector. In this interaction, the spatial gradient of the ALP field acts as an oscillating pseudomagnetic field on nuclear spins. When the Larmor frequency of the sample is tuned to the ALP Compton frequency, this weak drive can generate a transverse magnetization that is detected by the CASPER-Gradient-LowField apparatus.

The theoretical work developed the signal model and the data analysis framework needed for such a search. Two dark-matter scenarios were considered. For virialized Milky Way halo ALPs, the field is a stochastic superposition of many velocity modes, giving a finite coherence time and a linewidth set by the standard halo model. For Earth-bound ALPs, the field is instead described by bound states in the Earth's gravitational potential, leading to discrete eigenstates, longer coherence times, and a potentially higher dark matter density than that of the virialized halo.

The frequency-scanning analysis identified four signal regimes determined by the ordering of T_2^* , T_2 , and the ALP coherence time τ_a . In all long-measurement-duration regimes considered here, the coupling sensitivity scales only as $T_{\text{tot}}^{-1/4}$ with total measurement time. To maximize the sensitivity, the relaxation times T_2^* and T_2 should be maximized, regardless of whether a narrow or broad frequency band is being searched. The optimal frequency step size equals the sensitive bandwidth of a single measurement.

Experimentally, the thesis applied this framework to proof-of-principle measurements with thermally polarized methanol.

The December 23, 2022 Milky Way ALP search scanned a frequency band of $1.348\,570\text{ MHz} \pm 120\text{ Hz}$, corresponding to ALP masses $5.577\,237\text{ neV}/c^2 \pm 496\text{ feV}/c^2$. Three sequential scans were performed, with 100 s ALP-search acquisitions at each frequency step. The data were processed through sanity checks, Fourier transformation, lock-in response correction, Savitzky–Golay baseline removal, optimal filtering with the expected ALP lineshape, and normalized-power-excess thresholding. No statistically significant Milky Way ALP candidate survived the cross-scan

consistency checks. The resulting 95 % confidence-level constraint is

$$|g_{\text{ap}}| \lesssim 10^{-2} \text{ GeV}^{-1} \quad (5.1)$$

over the scanned mass range. This narrow-band search serves as a commissioning measurement for CASPER-Gradient-LowField and, to the best of our knowledge, is the first dedicated laboratory search in this mass range.

The December 14, 2022 Earth-bound ALP measurement used the same apparatus and sample but employed a longer measurement duration. It scanned $1.348\,575 \text{ MHz} \pm 24 \text{ Hz}$ with 1 h no-pulse acquisitions at each of eight scan steps, targeting the narrower linewidth and longer coherence time expected for Earth-bound ALP states. This frequency range corresponds to ALP masses $5.577\,258 \text{ neV}/c^2 \pm 99 \text{ feV}/c^2$. The analysis framework for this data set was structured analogously to that of the Milky Way search, but a one-bin statistic for unresolved signals replaced the optimal-filter convolution because the ALP linewidth is smaller than the resolution bandwidth. No statistically significant Earth-bound ALP candidate was found, and the resulting 95 % confidence-level constraint is approximately 10^{-9} GeV^{-1} over the scanned range.

The significance of these measurements is the demonstration of liquid-state NMR as an experimental method for searching for ALP-proton gradient coupling. The Milky Way ALP search establishes the operation of the CASPER-Gradient-LowField apparatus and its analysis procedure as a commissioning measurement [59]. This thesis extends that approach to search for Earth-bound ALPs. Together, these contributions provide an experimental and analytical basis for future searches with hyperpolarized samples and for interpreting their results under different ALP halo assumptions.

5.2 Outlook

The most direct path to improved sensitivity is higher nuclear-spin molar polarization. The thermally polarized methanol sample used here has a proton polarization of only approximately 1.8×10^{-7} , which limits the sensitivity. Hyperpolarized methanol, produced for example by brute-force prepolarization, parahydrogen-induced polarization, or dynamic nuclear polarization, could improve the projected coupling sensitivity by up to 6 orders of magnitude. Liquid ^{129}Xe polarized by spin-exchange optical pumping offers fractional polarization approaching unity together with a long T_2 . Recent progress in ^{129}Xe hyperpolarization and in the development of the temperature-controlled sample probe for the CASPER-Gradient-

HighField setup demonstrates the potential to improve the sensitivity of CASPER-Gradient-LowField.

Further gains should come from longer coherence in the sample and a quieter, more homogeneous apparatus. Improving the bias-field homogeneity from the present 10 ppm scale toward the projected 2 ppm level would increase both T_2^* and the fraction of spins resonant with a narrow ALP signal. The LF setup is also expected to benefit from an upgraded variable-temperature insert (VTI), which should improve temperature stability and operating flexibility. Lower electronic noise, larger effective sample volume and improved pickup-coil performance would enhance the sensitivity. These improvements are complementary: polarization increases the available magnetization, while long relaxation times increase the accumulated ALP-induced tipping angle.

The Earth-bound ALP analysis opens a second direction for future work. Because bound states can have long coherence times and geometry-dependent gradients, the optimal statistic, signal normalization, and validation differ from those used for the Milky Way halo. Completing this analysis requires finalizing the state-population assumptions, the reference signal power, the recovery efficiency, and the candidate consistency criteria. The transient jumps seen in the one-hour records should also be catalogued systematically. They are likely instrumental in origin, but the same tools may become useful for searches for transient exotic signals such as ALP domain walls.

Finally, the numerical and statistical framework developed here can be extended beyond the specific methanol measurements. The axionbloch simulations provide a way to test Bloch dynamics for realistic relaxation, field inhomogeneity, hyperpolarized T_1 decay, and stochastic ALP fields. Together with the broader frequency reach of CASPER-Gradient-LowField and the HighField apparatus, these tools prepare CASPER for wider scans of ALP parameter space and for comparisons between different dark-matter models, spin species, and coupling structures. One interesting direction is the dual-NMR-sample concept, which could expand the experimental reach to correlated searches for axions and other new-physics signals such as dark photons [24].

Data and source code availability

In spirit of open science, the source of this dissertation, including $\text{\LaTeX}$ files, program scripts, and figures, is available in the GitHub repository of the CASPER Collaboration:

<https://github.com/CASPER-Collaboration/ZhangYuzhe-Dissertation>

The numerical simulation package `axionbloch` used in this thesis is available as open-source software [63]:

<https://github.com/Yuzhe98/AxionBloch>

Raw data and analysis scripts using the raw data are available upon reasonable request to the CASPER Collaboration.

The outliers appeared in at least two independent scan steps with consistent frequencies and coupling strengths are regarded as ALP candidates. This appendix lists the ALP candidates identified in the Milky Way and Earth-halo searches.

Table A.1: Milky Way ALP candidates.

Index	Step index	Frequency (Hz)	Mass (neV/c ²)
1	5, 27	1348668.58	5.57764506
2	66, 86, 102	1348485.70	5.57688873
3	111, 117, 128	1348792.89	5.57815918
4	126, 140	1348872.72	5.57848936
5	7, 10	1348686.37	5.57771866
6	29, 95	1348769.02	5.57806045

Table A.2: Earth-halo ALP candidates.

Index	Step index	Frequency (Hz)	Mass (neV/c ²)
1	7	1348672.0739	5.5776595297
2	2	1348646.2182	5.5775525993
3	7	1348646.2294	5.5775526455
4	7	1348646.2375	5.5775526792
5	7	1348646.2472	5.5775527189
6	3	1348514.7177	5.5770087568
7	4	1348505.8949	5.5769722685

Index	Step index	Frequency (Hz)	Mass (neV/c ²)
8	5	1348505.3398	5.5769699730
9	5	1348505.3520	5.5769700234
10	5	1348505.8094	5.5769719150
11	5	1348505.8120	5.5769719257
12	5	1348505.8628	5.5769721358
13	5	1348505.9028	5.5769723014
14	5	1348505.9126	5.5769723419
15	5	1348505.9161	5.5769723564
16	5	1348595.9240	5.5773445990
17	5	1348595.9522	5.5773447157
18	5	1348595.9537	5.5773447217
19	5	1348595.9770	5.5773448182
20	5	1348595.9908	5.5773448753
21	5	1348596.0179	5.5773449876
22	5	1348596.0282	5.5773450299
23	5	1348646.2350	5.5775526685
24	5	1348646.2589	5.5775527675
25	6	1348505.8143	5.5769719353
26	6	1348505.8432	5.5769720548
27	6	1348505.8699	5.5769721653
28	6	1348505.8731	5.5769721784
29	6	1348505.8740	5.5769721821
30	6	1348505.8809	5.5769722106
31	6	1348505.8885	5.5769722423

Index	Step index	Frequency (Hz)	Mass (neV/c ²)
32	6	1348505.8896	5.5769722469
33	6	1348505.8963	5.5769722745
34	6	1348595.9210	5.5773445867
35	6	1348595.9736	5.5773448042
36	6	1348596.0053	5.5773449353
37	6	1348596.0214	5.5773450017
38	6	1348596.0231	5.5773450090

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